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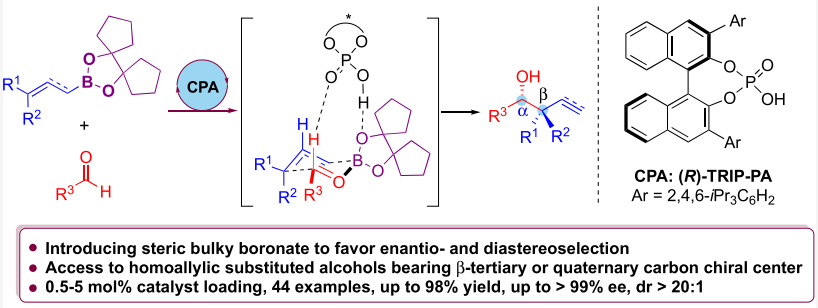
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Bi(cyclopentyl)diol-Derived Boronates in Highly Enantioselective Chiral Phosphoric Acid-Catalyzed Allylation, Propargylation, and Crotylation of Aldehydes

Jinping Yuan, Pankaj Jain, and Jon C. Antilla*

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ABSTRACT: In this study, we disclose the catalytic addition of bi(cyclopentyl)diol-derived boronates to aldehydes promoted by chiral phosphoric acids, allowing for the formation of enantioenriched homoallylic, propargylic, and crotylic alcohols (up to >99% enantiomeric excess (ee), diastereomeric ratio (dr) >20:1). These boronate substrates provided superior enantioselectivities, allowing for the reactions to proceed with low catalyst loading (0.5−5 mol %) and reduced reaction time (15 min at room temperature for aldehyde allylboration). A wide substrate scope was exhibited, and the novel boronates provided high enantiocontrol. Reactions with substituted allylboronates and aldehydes yielded vicinal stereogenic alcohols bearing *β*-tertiary or quaternary carbon centers. High enantio- and diastereoselectivities were found due to the closed six-membered chair-like transition state, with backbone modifications of the boronate and its interactions with the chiral phosphoric acid being the most likely contributing factor.

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1. INTRODUCTION

Asymmetric carbonyl allylation, propargylation, and crotylation represent important and powerful processes in stereoselective synthesis.[1] Optically active *β*-substituted alkanols with terminal alkynes/alkenes are found with high frequency in enantiospecific polyketide-derived natural product construction[2] (Scheme 1a), with methods development in this area therefore of great interest. Generally, methods for preparing chiral homoallyl/propargyl or crotylic alcohol systems rely on the addition of the nucleophilic allyl/allenyl reagent to achiral aldehydes for new carbon−carbon bond formation (Scheme 1b).[3,4] Hoffmann published the first enantioselective aldehyde allylation employing a camphor-derived allylboronic ester (1978).[4a] Subsequently, the labs of Brown,[4b] Roush,[4c] Cory,[4d] Leighton,[4e] Yamamoto,[4f] Aggarwal,[4g] Singaram,[4h] Denmark,[5a−c] and others[5d,4i] reported allyl-, propargyl-, and crotylations mediated by Lewis acids, modified chiral ligand/metal catalysts, or even chiral auxiliaries in the last 2 decades. Additionally, Hoveyda reported an efficient pathway for the asymmetric synthesis of homoallylic alcohols using chiral aminophenol-based reagents or the N-heterocyclic carbene (NHC)−Cu complex as catalysts.[6] Krische reported iridium or ruthenium-catalyzed asymmetric allyl-/crotyl-/propargylations via alcohol-mediated hydrogen autotransfer, which exploits nonmetallic allyl acetate, butadiene, and enyne pronucleophiles.[7] Despite enantioselectivities being high in some cases, the highly efficient route to produce enantiopure and vicinal stereogenic homoallyl/propargylic alcohols with a broad substrate scope under organocatalytic conditions is still attractive to many.

In 2010, triisopropylphenyl-substituted BINOL phosphoric acid, TRIP-PA[8] (**PA1**), was applied as a robust chiral catalyst in the allylboration of aldehydes by our group.[9] This method has resulted in significant use in synthesis[10] and has spurred the development of new methods by others.[11,12] Most of these aldehyde allylation and allenylation reactions utilize TRIP-PA

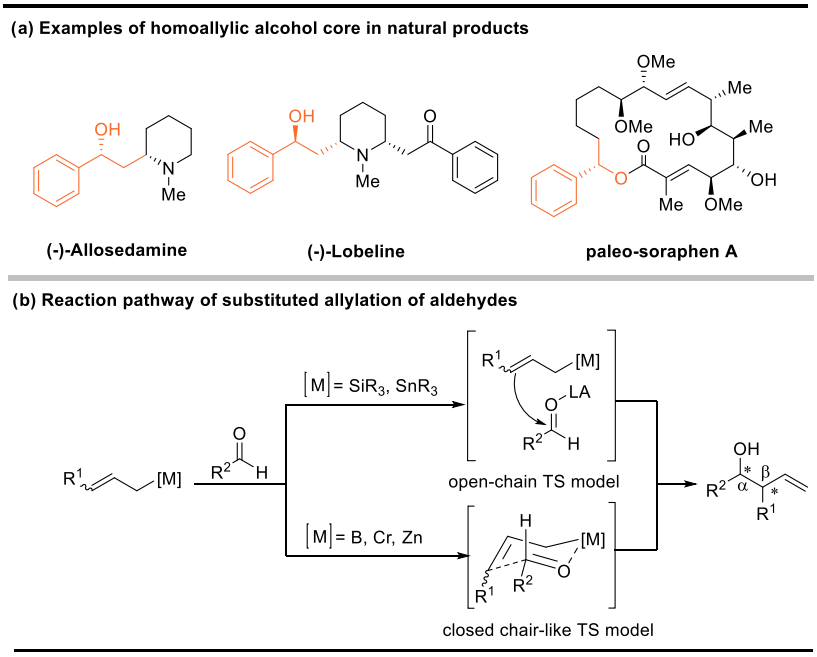
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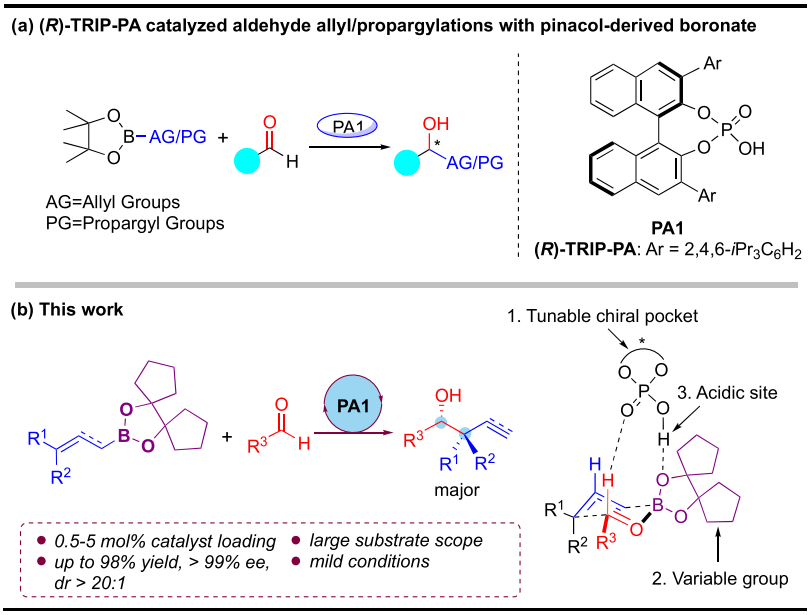
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Scheme 1. Background of Enantioselective Allylation of Aldehydes



Scheme 2. Chiral Phosphoric Acid (CPA)-Catalyzed Allyl-/Propargylations with Boronates and Aldehydes

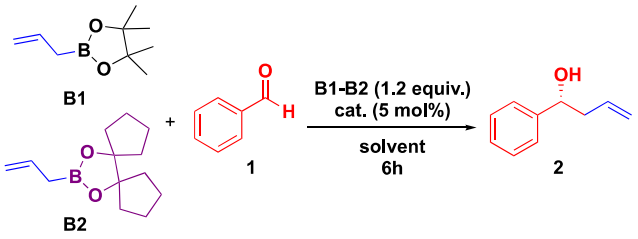


as the catalyst, along with boronates derived from pinacol to attain high enantiocontrol (Scheme 2a). However, a relatively high catalyst loading (20 mol %)[11a] or long reaction time (96 h for propargylation)[11a] was required in some cases.

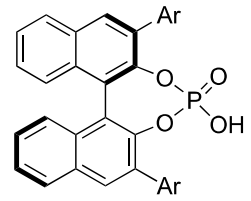
In catalytic synthesis, it has always been challenging to reduce catalyst quantity while retaining high enantio- and diastereoselectivity. In our initial insight into the allylation reaction mechanism, we proposed the aldehyde allylation with pinacol-derived boronate via (*R*)-TRIP-PA catalysis involving a closed six-membered chair-like transition state (TS), while the Lewis acid-catalyzed asymmetric allylation with allylsilanes and stannanes of aldehydes proceeded with an open-chain transition model (Scheme 1b).[5] In subsequent years after our initial paper, computational studies on (*R*)-TRIP-PA-catalyzed allyl/propargylation of pinacol boronates and benzaldehyde have been reported by Goodman and Houk independently.[13] Density functional theory (DFT) calculations suggested that the homoallyl/propargylic product's major isomer was generated by the bifunctional activation, which involves hydrogen bonding between boronate oxygen and the hydroxyl group of the catalyst with additional stabilization of the phosphoryl oxygen atom to the formyl hydrogen on the aldehyde. The enantioselectivity is believed to arise from steric interactions between bulky triisopropylphenyl groups on (*R*)-TRIP-PA and methyl groups on pinacol boronate for CPA-catalyzed allylation. Consequently, the enantioselectivity would be affected by the hindrance of the catalyst or boronate. Herein, we report a novel method of using CPA-catalyzed

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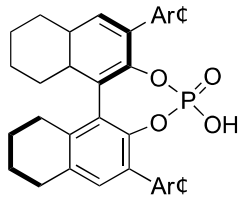
Table 1. Enantioselectivity with Modified Boronates[a]



Entry	Boronate	Catalyst	Solvent	Temp.	Conversion of **1**[b]	Ee
1	B1	PA1	toluene	rt	100%	93%
2	B1	PA2	toluene	rt	100%	93%
3	B1	PA2	cyclohexane	rt	100%	94%
4	B2	PA1	cyclohexane	rt	100%	99%
5	B2	PA2	cyclohexane	rt	100%	94%
6	B2	PA3	cyclohexane	rt	100%	77%
7	B2	PA1[c]	cyclohexane	rt	100%	**>99%** (**Method A**)
8	B2	PA1[d]	toluene	-30 °C	100%	**>99%** (**Method B**)



PA1: Ar = 2,4,6-*i*Pr$_3$C$_6$H$_2$
PA2: Ar = 2,4,6-cyclohexyl$_3$C$_6$H$_2$



PA3 Ar¢ = 2,4,6-*i*Pr$_3$C$_6$H$_2$

[a]Reaction conditions: all reactions were performed with **1** (0.10 mmol), **B1** or **B2** (0.12 mmol), 5 mol % catalyst, 50 mg 4 Å molecular sieve (MS), and 1 mL solvent. [b]Complete conversion was found in all cases. [c]**PA1** (2 mol %) was used, and the reaction was completed in 15 min. [d]**PA1** (2 mol %) was used.

bi(cyclopentyl)diol-derived boronates for allylation, propargylation, and crotylation of aldehydes based on mechanistic speculation. The use of backbone-modified boronates in the reaction not only reduces chiral catalyst loading under room temperature conditions but also high stereocontrol is retained. We also report additional examples of chiral homoallylic alcohols with β-tertiary or quaternary carbon centers in this study (Scheme 2b).

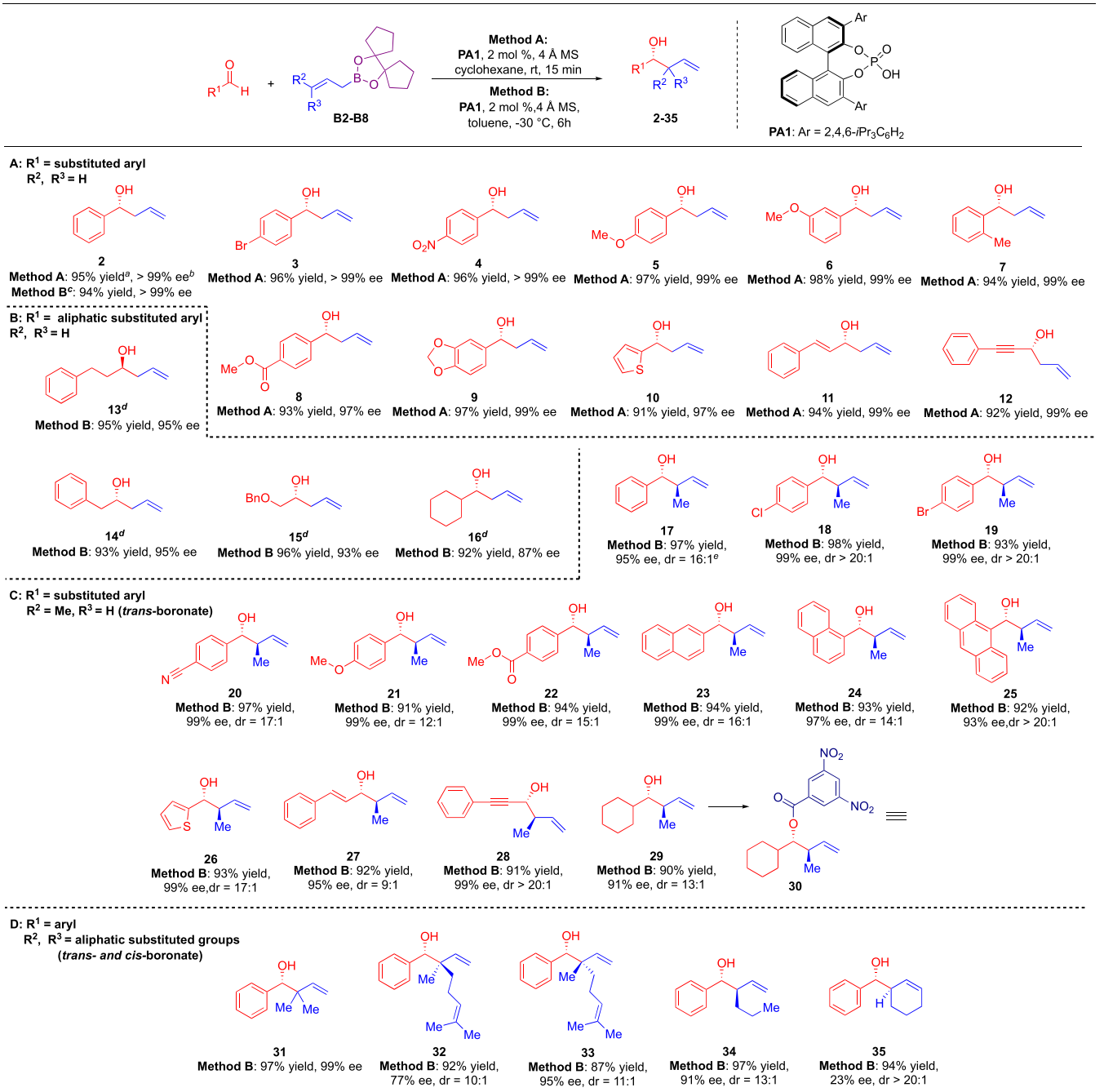
2. RESULTS AND DISCUSSION

The initial studies followed allylboration with benzaldehyde (Table 1). In our earlier report, allylation between pinacol-derived allylboronate **B1** and benzaldehyde catalyzed by 5 mol % **PA1** in toluene could achieve 93% enantiomeric excess (ee) at room temperature (entry 1).[8] In this investigation, we have rescreened different catalysts for allylboration, as some results published previously could be impacted by some metal impurities from silica gel. The results were different as all of the catalysts screened in the previous paper were not washed with HCl after purification by column chromatography on silica gel.[8] With 5 mol % acidified **PA1**, we found that the ee was 94%, which was similar to the 93% ee (see the Supporting Information for more details on catalyst rescreening). We then chose 5 mol % 2,4,6-cyclohexylphenyl-substituted BINOL phosphoric acid **PA2** as a bulkier catalyst than **PA1**, and the reaction achieved 93% ee at room temperature in toluene (entry 2). This enantioselectivity was equal to that attained by **PA1** with toluene as the solvent. Additionally, the enantioselectivity was 94% in cyclohexane at room temperature with 5 mol % **PA2** (entry 3).

Since the use of bulkier catalyst **PA2** did not have a significant influence on the enantioselectivity, we concentrated on modifying the backbone on the allylboronate. The reaction of bi(cyclopentyl)diol-derived allylic boronate **B2** and benzaldehyde successfully afforded **2** with 99% enantioselectivity at room temperature in cyclohexane in the presence of 5 mol % **PA1** catalysis (entry 4). Boronate **B2** was synthesized by reacting bi(cyclopentyl)diol with allylboronic acid. When boronate **B2** was used along with 5 mol % catalyst **PA2** at room temperature, 94% ee was obtained (entry 5). The reaction catalyzed by 5 mol % **PA3** gave only 77% ee in cyclohexane (entry 6). We further reduced **PA1** loading to 2 mol %, the reaction with **B2** at room temperature gave >99% ee with 100% conversion in less than 15 min (entry 7, method

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Scheme 3. Substrate Scope of Asymmetric-Substituted Allylboration of Aldehydes



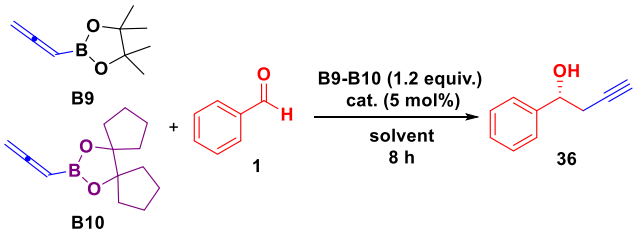
[a]Isolated yields. [b]The configurations were determined by chiral high-performance liquid chromatography (HPLC) analysis and optical rotation data in the literature. [c]0.5 mol % **PA1** employed at −30 °C. [d]**PA3** was used in **part B** for aliphatic aldehyde examples. [e]Determined by ^{1}H NMR analysis of crude products.

A). To test the efficiency of this aldehyde allylation method with **B2**, we attempted the reduced catalyst loading at lower temperatures. We found that >99% ee was obtained with only 2 mol % **PA1** at −30 °C in toluene (entry 8, method B). Usually, catalytic or noncatalytic asymmetric allylborations of aldehydes are carried out at −78 °C. To our knowledge, this study is the first report furnishing high selectivity's up to >99% ee at room temperature with only 2 mol % catalyst loading for enantioselective allylation of aldehydes.

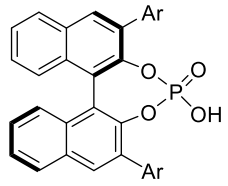
We then explored the substrate scope of the aldehyde allylboration (Scheme 3A). With method A, which utilized 2 mol % **PA1** at room temperature in cyclohexane, the reaction of a large range of aldehydes with electron-withdrawing and electron-donating groups at different positions on the aromatic ring with **B2** provided excellent yields and enantioselectivities (examples 2−9). Excellent enantioselectivities of 97 and 99% were also presented with heteroaromatic, and α,β-unsaturated aldehydes (examples 10 and 11, respectively). In addition, we synthesized homoallylic propargylic alcohol in 92% yield and with 99% ee (example 12). Surprisingly, we found that in the presence of only 0.5 mol % **PA1**, the reaction of **B2** and

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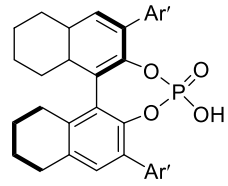
Table 2. Asymmetric Propargylation[a]



Entry	Boronate	Catalyst	Solvent	Temp.	Conversion of 1[b]	Ee
1	B9	PA1[c]	toluene	-20 °C	100%	91%
2	B9	PA1	toluene	rt	100%	77%
3	B9	PA1	cyclohexane	rt	100%	88%
4	B9	PA1	cyclohexane	10 °C	100%	92%
5	B9	PA2	cyclohexane	rt	100%	87%
6	B10	PA1	cyclohexane	rt	100%	**98%**
7	B10	PA2	cyclohexane	rt	100%	95%
8	B10	PA3	cyclohexane	rt	100%	74%



PA1: Ar = 2,4,6-*i*Pr$_3$C$_6$H$_2$
PA2: Ar = 2,4,6-cyclohexyl$_3$C$_6$H$_2$



PA3 Ar′ = 2,4,6-*i*Pr$_3$C$_6$H$_2$

[a]Reaction conditions: all propargylations were performed with **1** (0.10 mmol), **B9** or **B10** (0.12 mmol), 5 mol % catalyst, 50 mg 4 Å MS, and 1 mL solvent. [b]Complete conversion seen in all cases. [c]20 mol % **PA1**.

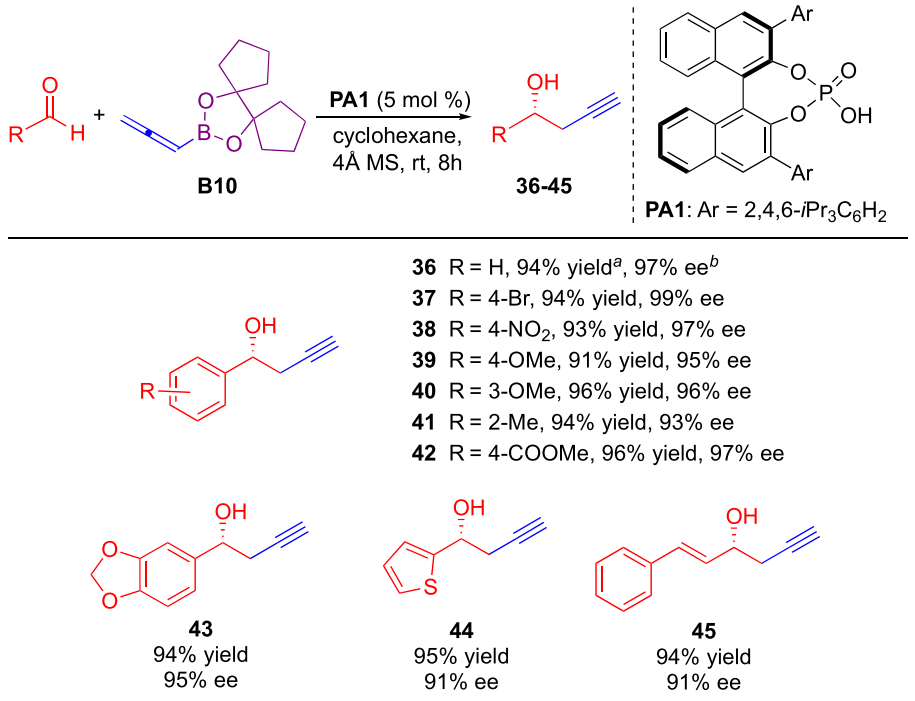
benzaldehyde at −30 °C in toluene proceeded with >99% ee (example 2, method B).

Subsequently, the scope of aliphatic aldehydes was investigated (Scheme 3B). As reported in our previous allylations, only moderate enantioselectivities with aliphatic aldehydes could be found even with larger catalyst quantities (*S*)-TRIP-PA (73−87% ee).[8] In this study, we found that when catalyzed by (*R*)-H8-TRIP-PA (**PA3**), the enantioselectivity of the reaction of hydrocinnamyl aldehyde and bi(cyclopentyl)diol-derived allylboronate **B2** was improved to 95% ee with method B (example 13) (see the Supporting Information for rescreening details on aliphatic aldehydes). Additional aliphatic-substituted aldehydes were also evaluated. Allylation with phenylacetaldehyde was shown to provide 93% yield with 95% ee (example 14). The reaction of benzyloxyacetaldehyde and **B2** gave 96% yield and 93% enantioselectivity, while pinacol boronate **B1** showed only 79% ee in our earlier report under similar conditions (example 15).[8] The reaction with cyclohexanecarboxaldehyde and **B2** gave 92% yield with 87% ee in the presence of 2 mol % **PA3** (example 16).

The crotylboration with aldehydes is a powerful strategy for *β*-methyl alkanol construction with two newly formed stereocenters. The crotylation process translates the geometry of the double bond on the crotylic reagent into relative syn/anti-products during C−C bond formation. When reacted with *trans*-crotyl boronate **B3**, a wide range of aldehydes with electron-rich and electron-poor groups on the aromatic ring was investigated for diastereo- and enantioselectivity with only 2 mol % **PA1** as the catalyst at −30 °C (Scheme 3C, method B). The products were major *anti*-homocrotylic alcohols with >90% yields, 95−99% enantioselectivities, and up to diastereomeric ratio (dr) >20:1 (examples 17−22). Additionally, we used heteroaryl, polycyclic aromatic, *α*,*β*-unsaturated, and aliphatic aldehydes to react with **B3**, providing excellent yields (90−94%) and enantio-/diastereoselectivities (91−99% ee, up to dr > 20:1, examples 23−29) for the anti-products. Furthermore, *γ*-substituted boronates **B4**−**B7** of both trans- and cis-isomers were well transformed into good yields (87−97%) and with high enantio- and diastereoselectivities (ee > 90%, see examples 31−34) into the corresponding *anti*/*syn*-alcohols with *β*-quaternary chiral carbon centers (Scheme 3D). Cyclohexene-derived boronate **B8** could be tolerated to give alcohol products with 94% yield and 23% ee (example 35). These enantioenriched homoallylic alcohols with extended carbon chains on the *β*-position could be used as asymmetric intermediates in synthesis, with this as a potential practical application. Notably, the absolute configuration of **29** was determined by its ester derivative **30**, and X-ray analysis

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Scheme 4. Substrate Scope of Asymmetric Propargylation of Aldehydes with B10



[a]Isolated yields. [b]The configurations were determined by chiral HPLC analysis and optical rotation data in the literature.

provided the absolute stereochemistry of this *anti*-ester with two adjacent chiral centers.[14]

We investigated the propargylboration of aldehydes, which has a slower reaction rate than that of aldehyde allylboration (Table 2). Previously, we reported that propargylation with allenyl pinacol boronate **B9** and benzaldehyde gave 91% ee in toluene at −20 °C with 20 mol % **PA1** catalysis (entry 1).[11a] In this optimization, our examination started by employing 5 mol % **PA1** in toluene with **B9** at room temperature, and 77% ee was achieved (entry 2). When cyclohexane was used as the solvent and the reaction catalyzed by 5 mol % **PA1** at room temperature with **B9**, the enantioselectivity was 88% (entry 3). This improvement in enantioselectivity indicated that cyclohexane was a superior solvent at room temperature. The reaction with **B9** and benzaldehyde catalyzed by 5 mol % **PA1** in cyclohexane gave 92% ee at 10 °C (entry 4).[11b] With the use of cyclohexane, propargylation of **B9** and benzaldehyde catalyzed by the relatively bulky substituted 5 mol % **PA2** gave 87% ee (entry 5). When bi(cyclopentyl)diol-derived allenyl ester **B10** and benzaldehyde reacted at room temperature along with cyclohexane under 5 mol % **PA1** catalysis, 98% ee was obtained (entry 6). The reaction of allenylboronate **B10** with the bulkier catalyst **PA2** gave 95% ee in cyclohexane (entry 7). The reaction of **B10** with **PA3** gave 74% enantioselectivity at room temperature (entry 8).

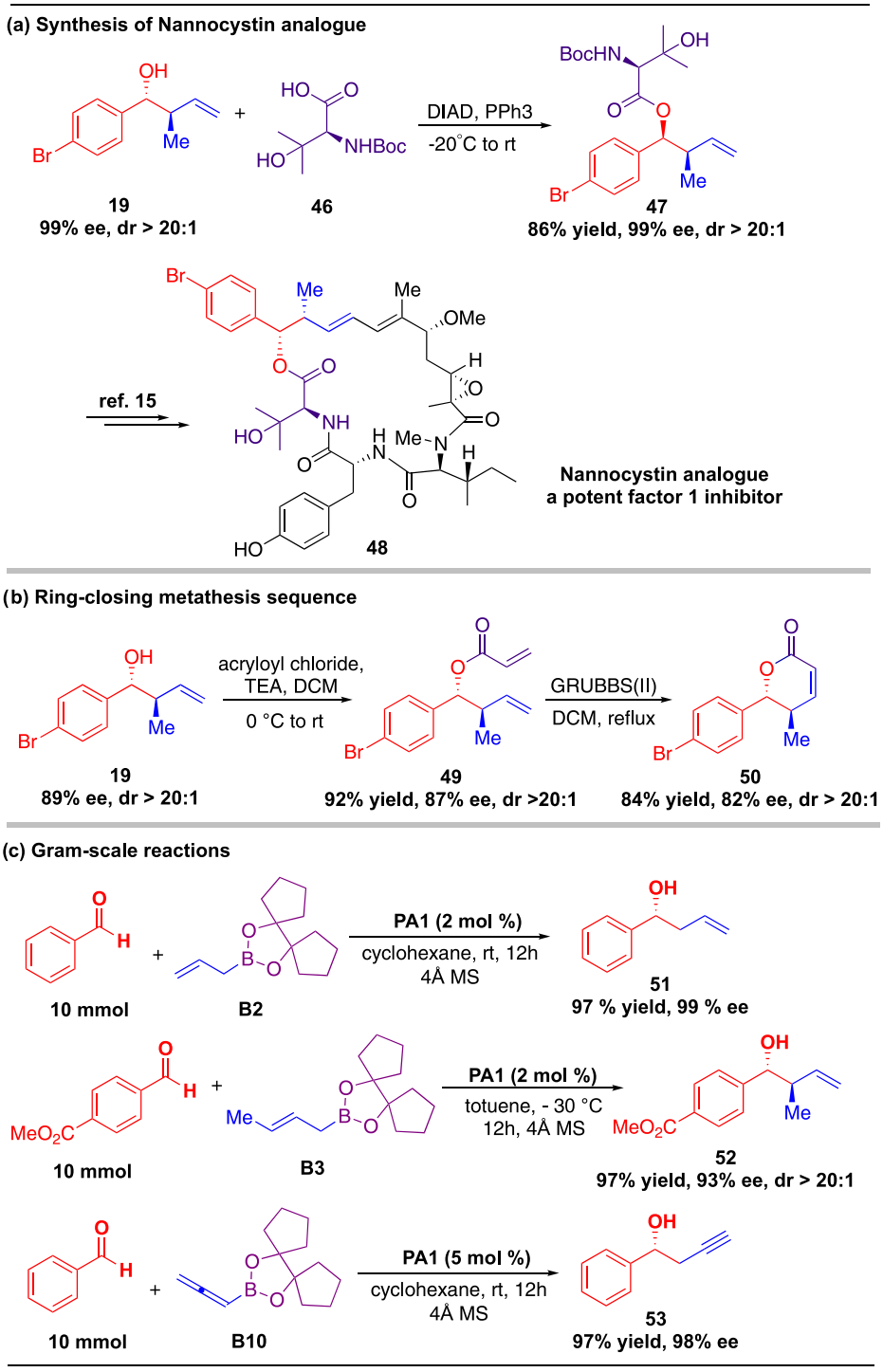
The substrate scope for the propargylation of **B10** and aldehydes was then examined with 5 mol % **PA1** at room temperature in cyclohexane (Scheme 4). Compared to earlier propargylation with **B9** and benzaldehyde,[10a,b] we found that with **B10**, the reaction times were shorter and superior enantioselectivities were achieved at room temperature. A broad range of homopropargylic alcohols was produced in high yields (91−96%) and with high enantioselectivities (91−99% ee) from different substituted aryl, heteroaromatic, and α,β-unsaturated aldehydes (examples 36−45).

To demonstrate the potential application of these asymmetric transformations, we applied the chiral homocrotylic alcohol to the synthesis of pharmaceutical intermediates (Scheme 5). Homocrotylic alcohol **19** was reacted with *N*-Boc-protected L-valine **46** to produce **47** with an identical 99% ee via Mitsunobu reaction. The furnished chiral amino acid **47** was a crucial chiral intermediate for the synthesis of nannocystin analogue **48**, which is a potent factor 1 inhibitor (Scheme 5a).[15] Moreover, diene-derived intermediate **49** was accessed by treatment with acryloyl chloride with little ee degradation from the starting alcohol. With Grubbs second-generation ruthenium catalyst under ring-closing metathesis conditions, pyranone **50** (83% ee, dr > 20:1) was obtained with a slight loss in ee (Scheme 5b). This procedure could be used for the synthesis of cyclic olefins of various ring sizes with the enantioenriched alcohols developed in our study as the starting material. Gram-scale reactions were carried out for practical utility tests. Excellent yields and enantio-/diastereoselectivities were realized for allylation (97 and 99% ee), crotylation (97 and 93% ee, dr > 20:1), and propargylation (97 and 98% ee) when conducted at the 10 mmol scale with benzaldehyde (Scheme 5c). The catalyst could be isolated in an 84% recovery rate from silica gel column chromatography with 5% methanol in dichloromethane (DCM) and then reacidified with 6 M HCl.

A general catalytic mechanism with the importance of steric interactions of the bicyclopentyl boronate and Brønsted acid catalyst was assumed (Scheme 6a). We believe that the reaction occurred through a bifunctional activation mode and underwent a chair-like six-membered ring cyclic TS that involved interaction of the pseudoaxial oxygen of boronate with the (*R*)-TRIP-PA hydroxyl hydrogen and stabilization of the formyl hydrogen on an aldehyde with the phosphoryl oxygen on the catalyst. This distortion of the "catalyst−boronate complex" indicated that a bulkier boronate ester

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Scheme 5. Synthetic Application Studies



group was consistent with increased stereoscopic repulsion. In this *re*-face favored attack model, bicyclopentyl boronate ester groups were directed toward the empty pocket of (*R*)-TRIP-PA, accounting for gauche interactions with the isopropyl-phenyls on the catalyst. The homoallylic intermediate **54** would easily hydrolyze to the corresponding alcohol (Scheme 6a). We conducted some benzaldehyde allylborations with 1,3-propanediol and ethylene glycol-derived allylboronates, in which the boronate ester group was relatively small (**B11** and **B12**). Lower enantioselectivities were observed and are shown in Scheme 6b (entries 1−4). Less than 50% ee was achieved at −78 °C, and even racemic products were formed at room temperature. The isopropanol-derived allylboronate (**B13**) was tested with benzaldehyde; enantioselectivity was 20% at room temperature and 32% at −78 °C (entries 5 and 6).

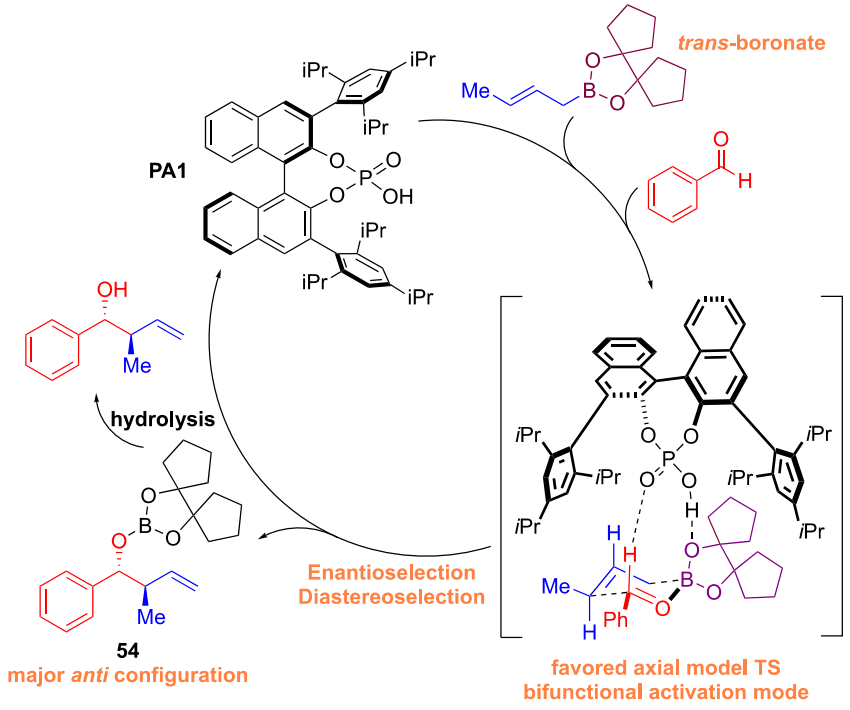
3. CONCLUSIONS

In this study, we identified bi(cyclopentyl)diol-derived allyl/allenyl boronates as superior nucleophiles for (*R*)-TRIP-PA-catalyzed allylation, propargylation, and crotylation of aldehydes. The reactions could proceed at room temperature or −30 °C with 0.5−5 mol % catalyst loading. Wide ranges of

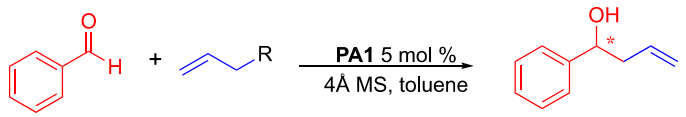
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Scheme 6. Catalyst Backbone: Impact on Selectivity

(a) Proposed mechanistic pathyway for crotylboration of benzaldehyde.



(b) Reaction of different allylboronates with benzaldehyde.[a]



1 B11-13

Entry	R	Temp (°C)	% Conversion of 1[b]	Ee (%)[c]	Major Isomer[d]
1	B11	rt	quant.	0	-
2		-78	quant.	22	*S*
3	B12	rt	quant.	0	-
4		-78	quant.	47	*R*
5	B13	rt	quant.	20	*S*
6		-78	quant.	32	*R*

[a]Reaction conditions: all reactions were performed with **1** (0.10 mmol), **B11**−**B13** (0.12 mmol), 5 mol % catalyst, 50 mg 4 Å MS, and 1 mL solvent. [b]Determined by ^{1}H NMR analysis. [c]The ee was determined by chiral HPLC analysis. [d]The configurations were determined by chiral HPLC and optical rotation data in the literature.

substrates were evaluated, and the reactions gave excellent yields and enantio- and diastereoselectivities (up to >99% ee, dr > 20:1). A bifunctional activation mode was assumed, and X-ray crystallography of a fully characterized *anti*-homoallylic alcohol confirmed the stereochemistry. These methods could be used in the preparations of relevant enantiopure pharmaceutical and natural product intermediates. Further expansion of substituted allylboration and propargylboration of aliphatic aldehydes are currently under investigation and will be reported in due course.

4. EXPERIMENTAL SECTION

4.1. General. All reactions were carried out in flame-dried screw-cap test tubes and were allowed to proceed under a dry argon atmosphere with magnetic stirring. Toluene was purified by passing through a column of activated alumina under a dry argon atmosphere.

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Anhydrous cyclohexane was purchased from Acros and directly used. Aldehydes were purchased from Adamas-β and were distilled prior to use if required. Thin-layer chromatography (TLC) was performed on Yantai TLC plates (silica gel 60 F_{254}). Visualization was accomplished by UV light (256 nm), with the combination of ceric ammonium molybdate or potassium permanganate as the indicator. Flash column chromatography was performed using Yantai silica gel (200−300 mesh). Enantiomeric excess (ee) was determined using a Varian Prostar HPLC system with a 210 binary pump and a 335 diode array detector; Agilent HPLC 1260 and Shimadzu HPLC LC-20AT. Optical rotations were performed on a Rudolph Research Analytical Autopol IV polarimeter (λ 589) using a 700 μL cell with a path length of 1 dm. ^{1}H NMR and ^{13}C NMR were recorded on a Bruker AVANCE IIITM HD NanoBAY (400 MHz) and Bruker AVANCE III (600 MHz) spectrometer with chemical shifts reported relative to tetramethylsilane (TMS). The high-resolution mass spectrometry (HRMS) data were measured on Thermo Fisher Q Exactive HF Orbitrap.

4.2. Preparation for the Bi(cyclopentyl)diol-Derived Boronate Substrates. *4.2.1. Synthesis of Diboron [1,1′-Bi(cyclopentane)]-1,1′-diol Ester.* In a flame-dried, two neck round-bottom flask, equipped with a magnetic stir bar, were placed 1.0 equiv tetrahydroxydiboron and 2.0 equiv [1,1′-bicyclopentyl]-1,1′-diol. Dry toluene was added, and the solution was stirred under argon at 110 °C for 24 h. Then, the reaction mixture was cooled to room temperature, the solvent was evaporated, and then concentrated in vacuo to produce a white solid in 91% yield. mp = 107−108 °C. Diboron [1,1′-bi(cyclopentane)]-1,1′-diol ester: ^{1}H NMR (600 MHz, $CDCl_3$) δ 1.87−1.51 (m, 32H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 92.0, 34.8, 22.1. HRMS (electrospray ionization (ESI)) calcd for $C_{20}H_{32}B_2O_4$ ([M + H]$^+$) m/z 359.2559, found 359.2555.

*4.2.2. Synthesis of Allylboronate **B2** from [1,1′-Bicyclopentyl]-1,1′-diol.* The allylboronate **B2** was synthesized from [1,1′-bicyclopentyl]-1,1′-diol (CAS 5181-75-9) following the literature procedure[16] for the synthesis of *B*-allyl-1,3,2-dioxaborinane. The boronate **B2** was obtained as a colorless oil in 83% yield. 12-Allyl-11,13-dioxa-12-boradispiro[4.0.46.35]tridecane (**B2**): ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.88−5.68 (m, 1H), 4.95−4.72 (m, 2H), 1.78 (d, J = 5.0 Hz, 2H), 1.73−1.41 (m, 16H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 133.3, 113.5, 92.1, 34.6, 21.9. HRMS (ESI) calcd for $C_{13}H_{21}BO_2$ ([M + Na]$^+$) m/z 243.1527, found 243.1520.

*4.2.3. Synthesis of Substituted Allylic Reagents **B3**−**B8**.* The noncommercially available allylic acetate substrates were generally prepared by the acetylation of the corresponding allylic alcohols or allylic chlorides. Following Morken's literature,[17] procedures were a little modified for the synthesis of **B3**−**B8**.

*4.2.3.1. (Z)-3,7-Dimethylocta-2,6-dien-1-yl Acetate (**AC1**).* The product was obtained by distillation (108−110 °C, 3 mmHg) to give a colorless oil. 85% yield. ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.29 (t, J = 7.2 Hz, 1H), 5.02 (t, J = 5.6 Hz, 1H), 4.48 (d, J = 7.3 Hz, 2H), 2.03 (t, J = 5.6 Hz, 4H), 1.97 (d, J = 2.2 Hz, 3H), 1.69 (s, 3H), 1.61 (s, 3H), 1.53 (s, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 170.0, 141.6, 131.1, 122.5, 118.1, 60.0, 31.1, 25.6, 24.6, 22.4, 20.0, 16.6. HRMS (ESI) calcd for $C_{12}H_{20}O_2$ ([M + Na]$^+$) m/z 219.1356, found 219.1357.

*4.2.3.2. (E)-3,7-Dimethylocta-2,6-dien-1-yl Acetate (**AC2**).* The product was obtained by distillation (109−111 °C, 3 mmHg) to give a colorless oil. 83% yield. ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.27 (t, J = 7.6 Hz, 1H), 5.01 (t, J = 6.8 Hz, 1H), 4.52 (d, J = 7.1 Hz, 2H), 2.07−1.98 (m, 4H), 1.98 (s, 3H), 1.63 (s, 3H), 1.61 (s, 3H), 1.53 (s, 3H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 170.1, 141.2, 130.8, 122.7, 117.2, 60.3, 38.5, 25.2, 24.6, 20.0, 16.6, 15.4. HRMS (ESI) calcd for $C_{12}H_{20}O_2$ ([M + H]$^+$) m/z 197.1534, found 197.1535.

*4.2.3.3. Cyclohex-2-en-1-yl Acetate (**AC3**).* The product was obtained by distillation (49−50 °C, 3 mmHg) to give a colorless oil. 89% yield. ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.88 (d, J = 17.3 Hz, 1H), 5.63 (d, J = 11.9 Hz, 1H), 5.18 (s, 1H), 2.04−1.88 (m, 5H), 1.80 (t, J = 15.3 Hz, 1H), 1.66 (d, J = 11.5 Hz, 2H), 1.59−1.53 (m, 1H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 169.6, 131.5, 124.7, 67.0, 27.3, 23.8, 20.3, 17.8. HRMS (ESI) calcd for $C_8H_{12}O_2$ ([M + Na]$^+$) m/z 163.0729, found 163.0728.

*4.2.3.4. (E)-Hex-2-en-1-yl Acetate (**AC4**).* The product was obtained by distillation (59−60 °C, 3 mmHg) to give a colorless oil. 92% yield. ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.75−5.64 (m, 1H), 5.56−5.42 (m, 1H), 4.43 (d, J = 6.6 Hz, 2H), 2.00−1.91 (m, 5H), 1.39−1.27 (m, 2H), 0.83 (td, J = 7.4, 1.4 Hz, 3H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 169.8, 135.3, 122.9, 64.2, 33.3, 21.0, 19.9, 12.6. HRMS (ESI) calcd for $C_8H_{14}O_2$ ([M + H]$^+$) m/z 143.1066, found 143.1066.

*4.2.3.5. (E)-12-(But-2-en-1-yl)-11,13-dioxa-12-oradispiro[4.0.4^6.3^5]tridecane (**B3**).* The product was obtained by flash column chromatography (pentane/ether, 50:1) as a colorless oil. 81% yield. ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.47−5.25 (m, 2H), 1.79−1.77 (m, 2H), 1.70−1.48 (m, 19H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 126.2, 125.0, 93.1, 35.7, 23.0, 18.0. HRMS (ESI) calcd for $C_{14}H_{23}BO_2$ ([M + H]$^+$) m/z 235.1860, found 235.1860.

*4.2.3.6. 12-(3-Methylbut-2-en-1-yl)-11,13-dioxa-12-boradispiro[4.0.4^6.3^5]tridecane (**B4**).* The product was obtained by flash column chromatography (pentane/ether, 50:1) as a colorless oil. 86% yield. ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.15 (ddd, J = 7.6, 4.6, 1.4 Hz, 1H), 1.78 (dd, J = 4.5, 2.4 Hz, 2H), 1.70−1.47 (m, 22H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 130.3, 117.7, 92.0, 34.6, 24.6, 22.0, 16.6. HRMS (ESI) calcd for $C_{15}H_{25}BO_2$ ([M + Na]$^+$) m/z 271.1839, found 271.1837.

*4.2.3.7. (E)-12-(3,7-Dimethylocta-2,6-dien-1-yl)-11,13-dioxa-12-boradispiro[4.0.4^6.3^5]tridecane (**B5**).* The product was obtained by flash column chromatography (pentane/ether, 50:1) and then distillation (176−177 °C, 3 mmHg) as a colorless oil. 81% yield. ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.61−4.93 (m, 2H), 2.05−1.88 (m, 4H), 1.84−1.38 (m, 27H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 134.9, 131.1, 124.5, 118.8, 93.0, 39.7, 35.6, 26.8, 25.7, 23.1, 17.7, 15.9. HRMS (ESI) calcd for $C_{20}H_{33}BO_2$ ([M + H]$^+$) m/z 317.2646, found 317.2642.

*4.2.3.8. (Z)-12-(3,7-Dimethylocta-2,6-dien-1-yl)-11,13-dioxa-12-boradispiro[4.0.4^6.3^5]tridecane (**B6**).* The product was obtained by flash column chromatography (pentane/ether, 50:1) and then distillation (174−175 °C, 3 mmHg) as a colorless oil. 78% yield. ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.68−4.90 (m, 2H), 2.15−1.84 (m, 4H), 1.81−1.34 (m, 27H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 135.0, 131.3, 124.6, 119.3, 92.9, 35.6, 31.8, 26.4, 25.7, 23.5, 23.0, 17.6. HRMS (ESI) calcd for $C_{20}H_{33}BO_2$ ([M + H]$^+$) m/z 317.2646, found 317.2642.

*4.2.3.9. (E)-12-(Hex-2-en-1-yl)-11,13-dioxa-12-boradispiro[4.0.4^6.3^5]tridecane (**B7**).* The product was obtained by flash column chromatography (pentane/ether, 50:1) and then distillation (155−157 °C, 3 mmHg) as a colorless oil. 81% yield. ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.56−5.11 (m, 2H), 1.89 (dt, J = 14.2, 7.1 Hz, 2H), 1.79−1.76 (m, 2H), 1.68−1.48 (m, 16H), 1.29 (dt, J = 14.6, 7.4 Hz, 2H), 0.80 (t, J = 7.4 Hz, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 130.5, 125.2, 93.1, 35.6, 34.8, 23.0, 22.7, 13.5. HRMS (ESI) calcd for $C_{16}H_{27}BO_2$ ([M + H]$^+$) m/z 263.2176, found 263.2170.

*4.2.3.10. 12-(Cyclohex-2-en-1-yl)-11,13-dioxa-12-boradispiro[4.0.4^6.3^5]tridecane (**B8**).* The product was obtained by flash column chromatography (pentane/ether, 50:1) and then distillation (90−91 °C, 3 mmHg) as a colorless oil. 72% yield. ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.71−5.44 (m, 2H), 2.05−1.85 (m, 2H), 1.83−1.69 (m, 5H), 1.66−1.45 (m, 16H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 127.9, 125.8, 93.0, 35.7, 35.5, 25.0, 24.1, 23.1, 22.5. HRMS (ESI) calcd for $C_{16}H_{25}BO_2$ ([M + H]$^+$) m/z 261.2020, found 261.2019.

*4.2.4. Synthesis of Allenylboronate **B10**.* The boronate **B10** was synthesized from allenylboronic acid pinacol ester following the literature procedure[18] for the synthesis of propanediolboronates (see the Supporting Information, a representative procedure for propanediolboronates (13, 16, 17): Pei, W.; Krauss, I. J. *J. Am. Chem. Soc.* **2011**, *133*, 18514). The boronate **B10** was obtained in 82% yield as a colorless oil. 12-(Propa-1,2-dien-1-yl)-11,13-dioxa-12-boradispiro-[4.0.4^6.3^5]tridecane (**B10**). ^{1}H NMR (400 MHz, $CDCl_3$) δ 4.91 (t, J = 7.0 Hz, 1H), 4.61 (d, J = 7.0 Hz, 2H), 1.94−1.54 (m, 16H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 217.8, 92.6, 68.8, 34.6, 22.0.

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HRMS (ESI) calcd for $C_{13}H_{19}BO_2$ ([M + Na]$^+$) m/z 241.1370, found 241.1362.

4.3. General Procedure for the Allylation of Aldehydes, Method A. A screw-cap reaction tube with a stir bar and 4 Å MS (50 mg) was evacuated, flame-dried, and back-filled with argon. To this tube were added the (*R*)-TRIP-PA catalyst **PA1** (2 mol %), freshly distilled aldehyde (0.1 mmol), and 1.0 mL of dry cyclohexane. Then, allylboronate **B2** (0.12 mmol) was added to the reaction mixture dropwise over 30 s. The reaction mixture was stirred for 15 min at room temperature (TLC shows complete conversion of the aldehyde) and then directly loaded on a silica gel column, and the crude product was purified by flash chromatography using ethyl acetate (EA) and hexane (1:9).

4.4. General Procedure for the Allylation of Aldehydes, Method B. A screw-cap reaction tube with a stir bar and 4 Å MS (50 mg) was evacuated, flame-dried, and back-filled with argon. To this tube were added the (*R*)-TRIP-PA catalyst **PA1** (2 mol %), freshly distilled aldehyde (0.1 mmol), and 1.0 mL of dry toluene. Then, the reaction mixture was then cooled to −30 °C, followed by the addition of allylboronate **B2** (0.12 mmol), dropwise over 30 s. The reaction mixture was stirred for 6 h at this temperature (TLC shows complete conversion of the aldehyde) and then directly loaded on a silica gel column, and the crude product was purified by flash chromatography using ethyl acetate and hexane (1:9).

*4.4.1. (R)-1-Phenyl-but-3-en-1-ol (**2**).* Colorless oil in 95% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 99/1, 0.7 mL/min), t_{major} = 28.61 min, t_{minor} = 33.53 min; ee = 99%. $[\alpha]_D^{24}$ = +57.23 (c = 1.07, benzene). The reported value[19] for the *R*-enantiomer (95% ee) is $[\alpha]_D$ = +56.5 (c = 1.0, benzene). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.31−7.18 (m, 5H), 5.82−5.64 (m, 1H), 5.15−5.02 (m, 2H), 4.67 (dd, J = 7.7, 5.3 Hz, 1H), 2.52−2.40 (m, 2H), 1.81 (s, 1H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 142.8, 133.4, 127.4, 126.5, 124.7, 117.4, 72.2, 42.8. HRMS (ESI) calcd for $C_{10}H_{12}O$ ([M + Na]$^+$) m/z 171.0780, found 171.0780.

*4.4.2. (R)-1-(4-Bromo-phenyl)-but-3-en-1-ol (**3**).* Colorless oil in 96% yield. Enantiomeric excess was determined by HPLC with a Chiralpak AD-H column (hexane/*i*PrOH = 98/2, 1.0 mL/min), t_{minor} = 13.29 min, t_{major}= 14.36 min; ee = 99%. $[\alpha]_D^{24}$ = +25.79 (c = 1.02, benzene). The reported value[19] for the *R*-enantiomer (96% ee) is $[\alpha]_D$ = +23.2 (c = 1.17, benzene). ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.40 (d, J = 8.4 Hz, 2H), 7.16 (d, J = 8.4 Hz, 2H), 5.76−5.59 (m, 1H), 5.12−5.05 (m, 2H), 4.63 (dd, J = 7.9, 5.0 Hz, 1H), 2.46−2.35 (m, 2H), 2.00 (s, 1H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 141.8, 132.9, 130.4, 126.5, 120.2, 117.8, 71.5, 42.8. HRMS (ESI) calcd for $C_{10}H_{11}BrO$ ([M + Na]$^+$) m/z 248.9885, found 248.9885.

*4.4.3. (R)-1-(4-Nitro-phenyl)-but-3-en-1-ol (**4**).* Colorless oil in 96% yield. Enantiomeric excess was determined by HPLC with a Chiralpak AD-H column (hexane/*i*PrOH = 98/2, 1.0 mL/min), t_{major} = 34.00 min, t_{minor} = 36.32 min; ee = 99%. $[\alpha]_D^{24}$ = +65.73 (c = 0.91, $CHCl_3$). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 8.13 (d, J = 8.7 Hz, 2H), 7.46 (d, J = 8.7 Hz, 2H), 5.72 (ddd, J = 17.0, 10.3, 7.2 Hz, 1H), 5.12 (dd, J = 13.2, 9.9 Hz, 2H), 4.80 (dd, J = 8.0, 4.6 Hz, 1H), 2.53−2.35 (m, 2H), 2.20 (s, 1H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 150.0, 146.2, 132.1, 125.5, 122.6, 118.6, 71.1, 42.9. HRMS (ESI) calcd for $C_{10}H_{11}NO_3$ ([M + H]$^+$) m/z 194.0811, found 194.0814.

*4.4.4. (R)-1-(4-Methoxy-phenyl)-but-3-en-1-ol (**5**).* Colorless oil in 97% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 98/2, 1.0 mL/min), t_{major} = 18.04 min, t_{minor}= 23.75 min; ee = 99%. $[\alpha]_D^{24}$ = +30.75 (c = 1.09, benzene). The reported value[20] for the *R*-enantiomer (95% ee) is $[\alpha]_D$ = +30.5 (c = 1.0, benzene). ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.21 (d, J = 8.6 Hz, 2H), 6.86−6.75 (m, 2H), 5.72 (ddt, J = 17.2, 10.2, 7.1 Hz, 1H), 5.17−4.96 (m, 2H), 4.61 (t, J = 6.5 Hz, 1H), 3.73 (s, 3H), 2.42 (t, J = 6.8 Hz, 2H), 1.95 (s, 1H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 158.0, 135.0, 133.6, 126.0, 117.1, 112.7, 71.9, 54.2, 42.7. HRMS (ESI) calcd for $C_{11}H_{14}O_2$ ([M + Na]$^+$) m/z 201.0886, found 201.0886.

*4.4.5. (R)-1-(3-Methoxy-phenyl)-but-3-en-1-ol (**6**).* Colorless oil in 98% yield. Enantiomeric excess was determined by HPLC with a Chiralpak AD-H column (hexane/*i*PrOH = 98/2, 1.0 mL/min), t_{minor} = 23.81 min, t_{major} = 25.75 min; ee = 99%. $[\alpha]_D^{24}$ = +55.09 (c = 1.12, benzene). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.19 (dd, J = 9.7, 6.4 Hz, 1H), 6.86 (dd, J = 4.5, 2.3 Hz, 2H), 6.75 (ddd, J = 8.3, 2.5, 0.8 Hz, 1H), 5.80−5.66 (m, 1H), 5.15−5.03 (m, 2H), 4.65 (dd, J = 7.9, 4.9 Hz, 1H), 3.75 (s, 3H), 2.44 (qt, J = 14.2, 7.1 Hz, 2H), 1.99 (s, 1H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 158.7, 144.6, 133.4, 128.4, 117.4, 117.1, 112.0, 110.3, 72.2, 54.2, 42.7. HRMS (ESI) calcd for $C_{11}H_{14}O_2$ ([M + H]$^+$) m/z 179.1066, found 179.1065.

*4.4.6. (R)-1-o-Tolyl-but-3-en-1-ol (**7**).* Colorless oil in 94% yield. Enantiomeric excess was determined by HPLC with a Chiralcel AD-H column (hexane/*i*PrOH = 95/5, 1.0 mL/min), t_{major} = 6.56 min, t_{minor}= 7.43min; ee = 99%. $[\alpha]_D^{24}$ = +70.77 (c = 1.2, benzene). The reported value[19] for the *R*-enantiomer (97% ee) is $[\alpha]_D$ = +75.5 (c = 1.0, benzene). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.40 (d, J = 7.5 Hz, 1H), 7.17−7.03 (m, 3H), 5.84−5.69 (m, 1H), 5.15−5.04 (m, 2H), 4.89 (dd, J = 8.2, 4.5 Hz, 1H), 2.41 (dddd, J = 26.1, 22.2, 10.2, 5.5 Hz, 2H), 2.26 (s, 3H), 1.90 (s, 1H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 140.9, 133.7, 133.3, 129.3, 126.2, 125.2, 124.1, 117.2, 68.6, 41.6, 18.0. HRMS (ESI) calcd for $C_{11}H_{14}O$ ([M + H]$^+$) m/z 163.1117, found 163.1116.

*4.4.7. (R)-Methyl 4-(1-hydroxybut-3enyl)benzoate (**8**).* Colorless oil in 93% yield. Enantiomeric excess was determined by HPLC with a Chiralpak AD-H column (hexane/*i*PrOH = 95/5, 1.0 mL/min), t_{major} = 13.17 min, t_{minor} = 14.99 min; ee = 97%. $[\alpha]_D^{24}$ = 28.08 (c = 1.07, benzene). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.94 (d, J = 8.3 Hz, 2H), 7.36 (d, J = 8.2 Hz, 2H), 5.72 (dddd, J = 11.4, 9.4, 7.7, 6.7 Hz, 1H), 5.10 (dd, J = 10.8, 6.4 Hz, 2H), 4.73 (dd, J = 8.0, 4.8 Hz, 1H), 3.84 (s, 3H), 2.51−2.36 (m, 2H), 2.16 (s, 1H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 165.9, 147.9, 132.8, 128.7, 128.2, 124.7, 118.0, 71.7, 51.0, 42.8. HRMS (ESI) calcd for $C_{12}H_{14}O_3$ ([M + H]$^+$) m/z 207.1016, found 207.1016.

*4.4.8. (R)-1-(Benzo[d][1,3]dioxol-5-yl)but-3-en-1-ol (**9**).* Colorless oil in 97% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 98/2, 1.0 mL/min), t_{major} = 21.55 min, t_{minor} = 26.97 min; ee = 99%. $[\alpha]_D^{24}$ = +35.79 c = 1.22, $CHCl_3$). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 6.74 (ddd, J = 29.4, 19.3, 4.7 Hz, 3H), 5.88 (s, 2H), 5.72 (ddt, J = 17.1, 10.2, 7.1 Hz, 1H), 5.12−5.03 (m, 2H), 4.58 (dd, J = 7.1, 5.9 Hz, 1H), 2.40 (td, J = 7.3, 2.2 Hz, 2H), 1.83 (s, 1H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 146.7, 145.9, 136.9, 133.3, 118.2, 117.3, 107.0, 105.3, 99.9, 72.1, 42.8. HRMS (ESI) calcd for $C_{11}H_{12}O_3$ ([M + Na]$^+$) m/z 215.0679, found 215.0678.

*4.4.9. (R)-1-Thiophen-2-yl-but-3-en-1-ol (**10**).* Colorless oil in 91% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OJ-H column (hexane/*i*PrOH = 99/1, 1.0 mL/min), t_{minor} = 26.12 min, t_{major} = 30.45 min; ee = 97%. $[\alpha]_D^{24}$ = +15.78 (c = 0.96, EtOH). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.19−7.15 (m, 1H), 6.92−6.86 (m, 2H), 5.75 (ddt, J = 17.2, 10.2, 7.1 Hz, 1H), 5.17−5.04 (m, 2H), 4.94−4.86 (m, 1H), 2.58−2.51 (m, 2H), 2.22 (s, 1H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 146.8, 132.8, 125.6, 123.5, 122.6, 117.7, 68.3, 42.7. HRMS (ESI) calcd for $C_8H_{10}OS$ ([M + Na]$^+$) m/z 177.0345, found 177.0344.

*4.4.10. (R,E)-1-Phenylhexa-1,5-dien-3-ol (**11**).* Colorless oil in 94% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 95/5, 1.0 mL/min), t_{major} = 13.13 min, t_{minor} = 22.64 min; ee = 99%. $[\alpha]_D^{24}$ = −11.83 (c = 1.03, Et_2O). The reported value[19] for the *R*-enantiomer (97% ee) is $[\alpha]_D$ = −12.3 (c = 1.0, Et_2O). ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.38 (d, J = 7.6 Hz, 2H), 7.31 (t, J = 7.6 Hz, 2H), 7.26−7.22 (m, 1H), 6.60 (d, J = 15.9 Hz, 1H), 6.24 (dd, J = 15.9, 6.3 Hz, 1H), 5.86 (ddt, J = 17.2, 10.2, 7.1 Hz, 1H), 5.22−5.14 (m, 2H), 4.36 (q, J = 5.7 Hz, 1H), 2.47−2.35 (m, 2H), 1.78 (s, 1H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 135.6, 133.0, 130.5, 129.3, 127.5, 126.6, 125.4, 117.5, 70.6, 41.0. HRMS (ESI) calcd for $C_{12}H_{14}O$ ([M + H]$^+$) m/z 175.1117, found 175.1118.

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*4.4.11. (R)-1-Phenylhex-5-en-1-yn-3-ol (**12**).* Colorless oil in 92% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 95/5, 1.0 mL/min), t_{major} = 9.21 min, t_{minor} = 26.06 min; ee = 99%. $[\alpha]_D^{24}$ = 23.46 (*c* = 0.21, $CHCl_3$). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.39−7.33 (m, 2H), 7.27−7.20 (m, 3H), 5.89 (ddt, *J* = 17.3, 10.2, 7.1 Hz, 1H), 5.20−5.12 (m, 2H), 4.59 (t, *J* = 6.0 Hz, 1H), 2.53−2.47 (m, 2H), 2.02 (s, 1H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 132.0, 130.6, 127.4, 127.2, 121.5, 118.1, 88.3, 84.2, 61.0, 41.2. HRMS (ESI) calcd for $C_{12}H_{12}O$ ([M + H]$^+$) *m*/*z* 173.0960, found 173.0961.

*4.4.12. (R)-1-Phenylhex-5-en-3-ol (**13**).* Colorless oil in 95% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 95/5, 1.0 mL/min), t_{major} = 8.77 min, t_{minor} = 13.95 min; ee = 95%. $[\alpha]_D^{24}$ = +25.9 (*c* = 0.89, benzene). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.21 (t, *J* = 7.6 Hz, 2H), 7.12 (dd, *J* = 15.7, 7.4 Hz, 3H), 5.84−5.66 (m, 1H), 5.13−5.01 (m, 2H), 3.61 (ddd, *J* = 12.2, 7.8, 4.6 Hz, 1H), 2.78−2.58 (m, 2H), 2.29−2.07 (m, 2H), 1.75−1.65 (m, 2H), 1.58 (s, 1H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 141.0, 133.5, 127.4, 127.3, 124.8, 117.3, 68.9, 41.0, 37.4, 31.0. HRMS (ESI) calcd for $C_{12}H_{16}O$ ([M + H]$^+$) *m*/*z* 177.1274, found 177.1274.

*4.4.13. (R)-1-Phenylpent-4-en-2-ol (**14**).* Colorless oil in 93% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 97/3, 0.5 mL/min), t_{minor} = 16.77 min, t_{major} = 22.07 min; ee = 95%. $[\alpha]_D^{24}$ = −14.03 (*c* = 1.0, $CHCl_3$). The reported value[21] for the *R*-enantiomer (97% ee) is $[\alpha]_D$= −14.24 (*c* = 0.65, $CHCl_3$). ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.32−7.26 (m, 2H), 7.23−7.18 (m, 3H), 5.92−5.75 (m, 1H), 5.20−5.05 (m, 2H), 3.85 (tt, *J* = 7.7, 4.8 Hz, 1H), 2.75 (ddd, *J* = 21.6, 13.6, 6.4 Hz, 2H), 2.35−2.13 (m, 2H), 1.81 (s, 1H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 137.3, 133.6, 128.4, 127.5, 125.4, 117.1, 70.6, 42.2, 40.1. HRMS (ESI) calcd for $C_{11}H_{14}O$ ([M + Na]$^+$) *m*/*z* 185.0937, found 185.0936.

*4.4.14. (R)-1-Benzyloxy-pent-4-en-2-ol (**15**).* Colorless oil in 96% yield. Enantiomeric excess was determined by HPLC with a Chiralpak AS-H column (hexane/*i*PrOH = 97/3, 0.5 mL/min), t_{minor} = 20.67 min, t_{major} = 23.57 min; ee = 93%. $[\alpha]_D^{24}$ = −2.76 (*c* = 1.15, $CHCl_3$). The reported value[22] for the *S*-enantiomer (53% ee) is $[\alpha]_D$ = +0.9 (*c* = 2.5, $CHCl_3$). ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.34−7.19 (m, 5H), 5.76 (ddt, *J* = 17.2, 10.2, 7.1 Hz, 1H), 5.05 (dd, *J* = 23.1, 5.9 Hz, 2H), 4.49 (s, 2H), 3.82 (ddd, *J* = 13.5, 6.7, 3.4 Hz, 1H), 3.45 (dd, *J* = 9.5, 3.4 Hz, 1H), 3.31 (dd, *J* = 9.5, 7.5 Hz, 1H), 2.20 (t, *J* = 6.8 Hz, 2H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 138.0, 134.2, 128.4, 127.8, 127.7, 117.7, 73.9, 73.4, 69.7, 37.9. HRMS (ESI) calcd for $C_{12}H_{16}O_2$ ([M + H]$^+$) *m*/*z* 193.1223, found 193.1223.

*4.4.15. (R)-1-Cyclohexyl-but-3-en-1-ol (**16**).* Colorless oil in 92% yield. Enantiomeric excess was determined by the formation of 3,5-dinitrobenzoate ester of the title compound followed by HLPC with a Chiralcel OD-H column (hexane/*i*PrOH = 95/5, 1.0 mL/min), t_{major} = 9.970 min, t_{minor} = 10.666 min; ee = 87%. $[\alpha]_D^{24}$ = +3.9 (*c* = 1.0, CH_2Cl_2). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.84−5.68 (m, 1H), 5.12−4.99 (m, 2H), 3.33 (ddd, *J* = 9.1, 5.8, 3.5 Hz, 1H), 2.33−2.19 (m, 1H), 2.06 (dt, *J* = 14.0, 8.5 Hz, 1H), 1.79 (d, *J* = 12.8 Hz, 1H), 1.65 (dd, *J* = 53.1, 2.6 Hz, 4H), 1.50 (s, 1H), 1.20−0.92 (m, 6H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 134.4, 116.9, 73.7, 42.0, 37.8, 28.0, 27.1, 25.5, 25.2, 25.1. HRMS (ESI) calcd for $C_{10}H_18O$ ([M + Na]$^+$) *m*/*z* 177.1250, found 177.1250.

4.5. General Procedure for the Substituted Allylation of Benzaldehyde with B3−B8. A screw-cap reaction tube with a stir bar and 4 Å MS (50 mg) was evacuated, flame-dried, and back-filled with argon. To this tube were added the (*R*)-TRIP-PA catalyst **PA1** (2 mol %), freshly distilled benzaldehyde (0.10 mmol), and 1.0 mL of dry toluene as a solvent. The reaction mixture was then cooled to −30 °C, followed by the addition of boronate **B3−B8** (0.12 mmol), dropwise over 30 s. The mixture was stirred 6 h at this temperature. Then, 1 mL of 1 M HCl was added and the reaction mixture was stirred for 15 min. Proton NMR of the crude mixture was collected for dr analysis, and then the product was purified by flash chromatography using ethyl acetate and hexanes (1:9).

*4.5.1. (1R,2R)-2-Methyl-1-phenylbut-3-en-1-ol (**17**).* Colorless oil in 97% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 90/10, 1.0 mL/min), t_{minor} = 5.297 min, t_{major} = 6.192 min; ee = 95%. $[\alpha]_D^{25}$ = +92.57 (*c* = 1.05, $CHCl_3$). The reported value[12b] for the *R*,*R*-enantiomer (99% ee) is $[\alpha]_D^{26}$ = +93.18 (*c* = 1.09, $CHCl_3$). ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.34−7.19 (m, 5H), 5.80−5.68 (m, 1H), 5.13 (t, *J* = 13.2 Hz, 2H), 4.29 (d, *J* = 7.9 Hz, 1H), 2.46−2.37 (m, 1H), 2.09 (s, 1H), 0.80 (d, *J* = 6.8 Hz, 3H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 141.4, 139.6, 127.2, 126.6, 125.8, 115.8, 76.8, 45.3, 15.5. HRMS (ESI) calcd for $C_{11}H_{14}O$ ([M + Na]$^+$) *m*/*z* 185.0937, found 185.0943.

*4.5.2. (1R,2R)-1-(4-Chlorophenyl)-2-methylbut-3-en-1-ol (**18**).* Colorless oil in 98% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 90/10, 1.0 mL/min), t_{minor} = 5.166 min, t_{major} = 5.455 min; ee = 99%. $[\alpha]_D^{25}$ = +22.01 (*c* = 0.15, MeOH). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.30 (dd, *J* = 28.5, 8.5 Hz, 4H), 5.77 (ddd, *J* = 17.0, 10.7, 8.4 Hz, 1H), 5.25−5.16 (m, 2H), 4.34 (d, *J* = 7.8 Hz, 1H), 2.49−2.37 (m, 1H), 2.17 (s, 1H), 0.87 (d, *J* = 6.8 Hz, 3H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 140.9, 140.1, 133.3, 128.3, 128.2, 117.2, 77.1, 46.3, 16.4. HRMS (ESI) calcd for $C_{11}H_{13}ClO$ ([M + Na]$^+$) *m*/*z* 219.0547, found 219.0548.

*4.5.3. (1R,2R)-1-(4-Bromophenyl)-2-methylbut-3-en-1-ol (**19**).* Colorless oil in 93% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 99/1, 1.0 mL/min), t_{minor} = 20.299 min, t_{major} = 20.823 min; ee = 99%. $[\alpha]_D^{25}$ = +39.78 (*c* = 0.85, MeOH). The absolute configuration was determined by analogy. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.40 (d, *J* = 8.4 Hz, 2H), 7.14 (d, *J* = 8.3 Hz, 2H), 5.76−5.61 (m, 1H), 5.17−5.07 (m, 2H), 4.25 (d, *J* = 7.7 Hz, 1H), 2.40−2.29 (m, 1H), 2.12 (s, 1H), 0.80 (d, *J* = 6.8 Hz, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 140.3, 139.1, 130.3, 127.5, 120.4, 116.3, 76.1, 45.3, 15.3. HRMS (ESI) calcd for $C_{11}H_{13}BrO$ ([M + Na]$^+$) *m*/*z* 263.0042, found 263.0043.

*4.5.4. 4-((1R,2R)-1-Hydroxy-2-methylbut-3-en-1-yl)benzonitrile (**20**).* Colorless oil in 97% yield. Enantiomeric excess was determined by HPLC with a Chiralpak AD-H column (hexane/*i*PrOH = 95/5, 1.0 mL/min), t_{major} = 14.132 min, t_{minor} = 20.221 min; ee = 99%. $[\alpha]_D^{25}$ = +15.39 (*c* = 1.50, MeOH). The absolute configuration was determined by analogy. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.59−7.54 (m, 2H), 7.38 (d, *J* = 8.1 Hz, 2H), 5.66 (ddd, *J* = 17.1, 10.4, 8.2 Hz, 1H), 5.16−5.07 (m, 2H), 4.38 (d, *J* = 7.1 Hz, 1H), 2.36 (dt, *J* = 14.2, 7.1 Hz, 1H), 2.27 (d, *J* = 2.4 Hz, 1H), 0.84 (d, *J* = 6.8 Hz, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 146.7, 138.3, 131.0, 126.5, 117.8, 116.8, 110.3, 75.9, 45.3, 15.2. HRMS (ESI) calcd for $C_{12}H_{13}NO$ ([M + Na]$^+$) *m*/*z* 210.0889, found 210.0880.

*4.5.5. (1R,2R)-1-(4-Methoxyphenyl)-2-methylbut-3-en-1-ol (**21**).* Colorless oil in 91% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OJ-H column (hexane/*i*PrOH = 95/5, 1.0 mL/min), t_{major} = 17.316 min, t_{minor} = 20.829 min; ee = 99%. $[\alpha]_D^{25}$ = +42.4 (*c* = 1.5, $CDCl_3$). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.26−7.24 (m, 2H), 6.90−6.85 (m, 2H), 5.81 (ddd, *J* = 17.2, 10.3, 8.2 Hz, 1H), 5.25−5.09 (m, 2H), 4.30 (d, *J* = 8.0 Hz, 1H), 3.80 (s, 3H), 2.46 (dt, *J* = 14.3, 7.1 Hz, 1H), 1.92 (s, 1H), 0.85 (d, *J* = 6.8 Hz, 3H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 159.1, 140.9, 134.6, 127.9, 116.5, 113.6, 77.5, 55.2, 46.3, 16.5. HRMS (ESI) calcd for $C_{12}H_{16}O_2$ ([M + Na]$^+$) *m*/*z* 215.11033, found 215.1035.

*4.5.6. Methyl 4-((1R,2R)-1-Hydroxy-2-methylbut-3-en-1-yl)-benzoate (**22**).* Colorless oil in 94% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 95/5, 1.0 mL/min), t_{major} = 13.187 min, t_{minor} = 18.128 min; ee = 99%. $[\alpha]_D^{25}$ = +25.2 (*c* = 0.95, MeOH). The absolute configuration was determined by analogy. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.96−7.92 (m, 2H), 7.33 (d, *J* = 8.2 Hz, 2H), 5.69 (ddd, *J* = 17.8, 9.7, 8.2 Hz, 1H), 5.19−5.03 (m, 2H), 4.37 (dd, *J* = 7.5, 2.5 Hz, 1H), 3.84 (s, 3H), 2.41 (dt, *J* = 14.6, 7.2 Hz, 1H), 2.21 (d, *J* = 1.4 Hz, 1H), 0.82 (d, *J* = 6.8 Hz, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 165.8, 146.5, 138.9, 128.5, 128.4, 125.8, 116.3, 51.1, 45.3, 15.3. HRMS (ESI) calcd for $C_{13}H_{16}O_3$ ([M + Na]$^+$) *m*/*z* 243.0992, found 243.0990.

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*4.5.7. (1R,2R)-2-Methyl-1-(naphthalen-2-yl)but-3-en-1-ol (**23**).* Colorless oil in 94% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 90/10, 1.0 mL/min), t_{minor} = 11.641 min, t_{major} = 12.893 min; ee = 99%. $[\alpha]_D^{25}$ = +19.04 (*c* = 1.14, MeOH). The absolute configuration was determined by analogy. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.75 (dd, *J* = 16.0, 13.9 Hz, 4H), 7.45−7.35 (m, 3H), 5.78 (ddd, *J* = 17.3, 10.2, 8.2 Hz, 1H), 5.21−5.11 (m, 2H), 4.46 (d, *J* =7.8 Hz, 1H), 2.60−2.47 (m, 1H), 2.19 (s, 1H), 0.83 (d, *J* = 6.8 Hz, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 139.5, 138.8, 132.1, 132.0, 127.0, 126.9, 126.6, 125.0, 124.9, 124.7, 123.6, 116.0, 76.9, 45.2, 15.6. HRMS (ESI) calcd for $C_{15}H_{16}O$ ([M + Na]$^+$) *m*/*z* 235.1093, found 235.1088.

*4.5.8. (1R,2R)-2-Methyl-1-(naphthalen-1-yl)but-3-en-1-ol (**24**).* Colorless oil in 93% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 90/10, 1.0 mL/min), t_{minor} = 8.326 min, t_{major} = 17.596 min; ee = 97%. $[\alpha]_D^{25}$ = +25.38 (*c* = 0.84, MeOH). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 8.15 (d, *J* = 8.4 Hz, 1H), 7.86−7.83 (m, 1H), 7.75 (t, *J* = 7.9 Hz, 1H), 7.54 (d, *J* = 7.1 Hz, 1H), 7.50−7.41 (m, 3H), 5.86 (ddd, *J* = 16.9, 10.7, 7.8 Hz, 1H), 5.18−5.12 (m, 3H), 2.80 (h, *J* = 6.9 Hz, 1H), 2.28 (s, 1H), 0.94 (d, *J* = 6.9 Hz, 3H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 140.2, 138.4, 133.9, 131.0, 128.9, 128.1, 125.8, 125.4, 125.2, 124.5, 123.6, 116.7, 74.8, 45.2, 17.1. HRMS (ESI) calcd for $C_{15}H_{16}O$ ([M + Na]$^+$) *m*/*z* 235.1093, found 235.1084.

*4.5.9. (1R,2R)-1-(Anthracen-9-yl)-2-methylbut-3-en-1-ol (**25**).* Yellow solid in 92% yield. mp = 121−122 °C. Enantiomeric excess was determined by HPLC with a Chiralpak AD-H column (hexane/*i*PrOH = 95/5, 1.0 mL/min), t_{minor} = 20.571 min, t_{major} = 23.962 min; ee = 93%. $[\alpha]_D^{25}$ = +12.59 (*c* = 2.25, MeOH). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 8.57 (s, 1H), 8.31 (d, *J* = 13.5 Hz, 2H), 7.91 (d, *J* = 8.2 Hz, 2H), 7.52−7.26 (m, 4H), 6.04 (ddd, *J* = 17.4, 10.1, 8.3 Hz, 1H), 5.85 (d, *J* = 10.0 Hz, 1H), 5.34 (d, *J* = 17.2 Hz, 1H), 5.24 (dd, *J* = 10.2, 1.2 Hz, 1H), 3.36−3.24 (m, 1H), 2.41 (s, 1H), 0.56 (d, *J* = 6.8 Hz, 3H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 141.0, 131.7, 130.6, 129.2, 128.3, 127.3, 124.5, 123.7, 115.6, 72.6, 44.4, 16.3. HRMS (ESI) calcd for $C_{19}H_{18}O$ ([M + Na]$^+$) *m*/*z* 285.1250, found 285.1249.

*4.5.10. (1R,2R)-2-Methyl-1-(thiophen-2-yl)but-3-en-1-ol (**26**).* Colorless oil in 93% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OJ-H column (hexane/*i*PrOH = 96/4, 1.0 mL/min), t_{minor} = 10.043 min, t_{major} = 10.569 min; ee = 99%. $[\alpha]_D^{25}$ = +11.23 (*c* = 0.77, MeOH). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.27−7.26 (m, 1H), 6.99−6.95 (m, 2H), 5.83 (ddd, *J* = 17.3, 10.3, 8.1 Hz, 1H), 5.26−5.18 (m, 2H), 4.66 (d, *J* = 7.7 Hz, 1H), 2.55 (dt, *J* = 14.6, 7.3 Hz, 1H), 2.25 (s, 1H), 0.96 (d, *J* = 6.8 Hz, 3H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 146.3, 140.0, 126.4, 124.7, 117.2, 73.9, 46.6, 16.5. HRMS (ESI) calcd for $C_9H_{12}OS$ ([M + Na]$^+$) *m*/*z* 191.0501, found 191.0498.

*4.5.11. (3S,4R,E)-4-Methyl-1-phenylhexa-1,5-dien-3-ol (**27**).* Colorless oil in 92% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 90/10, 1.0 mL/min), t_{major} = 7.074 min, t_{minor} = 9.823 min; ee = 95%. $[\alpha]_D^{25}$ = +13.90 (*c* = 1.09, MeOH). The absolute configuration was determined by analogy. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.39 (dd, *J* = 9.3, 2.2 Hz, 2H), 7.32 (dd, *J* = 10.2, 4.8 Hz, 2H), 7.24 (dd, *J* = 5.8, 3.6 Hz, 1H), 6.65−6.53 (m, 1H), 6.21 (dd, *J* = 15.9, 7.0 Hz, 1H), 5.87−5.75 (m, 1H), 5.23−5.17 (m, 1H), 5.16 (d, *J* = 0.8 Hz, 1H), 4.05 (t, J = 6.5 Hz, 1H), 2.37 (dd, *J* = 13.9, 7.7 Hz, 1H), 1.68 (s, 1H), 1.07 (t, *J* = 4.9 Hz, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 140.1, 136.7, 131.7, 130.2, 128.5, 127.6, 126.5, 116.6, 76.1, 44.6, 16.0. HRMS (ESI) calcd for $C_{13}H_{16}O$ ([M + Na]$^+$) *m*/*z* 211.1093, found 211.1086.

*4.5.12. (3R,4R)-4-Methyl-1-phenylhex-5-en-1-yn-3-ol (**28**).* Colorless oil in 91% yield. Enantiomeric excess was determined by HPLC with a Chiralpak AS-H column (hexane/*i*PrOH = 98/2, 1.0 mL/min), t_{major} = 12.498 min, t_{minor} = 15.472 min; ee = 99%. $[\alpha]_D^{25}$ = +15.26 (*c* = 0.55, MeOH). The absolute configuration was determined by analogy. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.46−7.33 (m, 2H), 7.31−7.19 (m, 3H), 5.91−5.72 (m, 1H), 5.13 (ddt, *J* = 6.4, 2.5, 1.6 Hz, 2H), 4.37 (t, *J* = 5.7 Hz, 1H), 2.56−2.43 (m, 1H), 1.96 (d, *J* = 5.4 Hz, 1H), 1.13 (d, *J* = 6.8 Hz, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 138.2, 130.7, 127.4, 127.2, 121.5, 115.8, 87.4, 84.7, 65.6, 43.5, 14.3. HRMS (ESI) calcd for $C_{13}H_{14}O$ ([M + Na]$^+$) *m*/*z* 209.0937, found 209.0932.

*4.5.13. (1S,2R)-1-Cyclohexyl-2-methylbut-3-en-1-ol (**29**).* Colorless oil in 90% yield. Enantiomeric excess was determined according to its ester derivative 30. ee = 91%. $[\alpha]_D^{25}$ = +19.48 (*c* = 0.55, $CHCl_3$). The reported value[23] for the *S*,*R*-enantiomer (98% ee) is $[\alpha]_D^{24}$ = +18.4 (*c* = 1.29, $CHCl_3$). ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.84−5.71 (m, 1H), 5.17−5.02 (m, 2H), 3.10 (t, *J* = 5.8 Hz, 1H), 2.44−2.32 (m, 1H), 1.84−1.73 (m, 3H), 1.68−1.60 (m, 2H), 1.48 (s, 1H), 1.40 (dddd, *J* = 11.7, 8.9, 6.0, 2.7 Hz, 1H), 1.27−1.06 (m, 5H), 1.03 (d, *J* = 6.9 Hz, 3H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 139.3, 115.1, 77.8, 39.5, 39.2, 29.0, 25.9, 25.4, 25.4, 25.1, 15.9. HRMS (ESI) calcd for $C_{11}H_{20}O$ ([M + Na]$^+$) *m*/*z* 191.1406, found 191.1399.

*4.5.14. (1S,2R)-1-Cyclohexyl-2-methylbut-3-en-1-yl 3,5-dinitrobenzoate (**30**).* White solid in 71% yield. mp = 68−69 °C. Enantiomeric excess was determined by HPLC with a Chiralcel IB N-5 column (hexane/*i*PrOH = 99/1, 1.0 mL/min), t_{major} = 14.145 min, t_{minor} = 15.182 min; ee = 91%. $[\alpha]_D^{25}$ = +6.41 (*c* = 0.45, MeOH). ^{1}H NMR (600 MHz, $CDCl_3$) δ 9.16 (t, *J* = 2.1 Hz, 1H), 9.07 (d, *J* = 2.1 Hz, 2H), 5.73 (ddd, *J* = 17.1, 10.2, 8.6 Hz, 1H), 5.03−4.95 (m, 3H), 2.72−2.53 (m, 1H), 1.72−1.59 (m, 6H), 1.23−1.08 (m, 5H), 0.99 (d, *J* = 6.9 Hz, 3H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 161.3, 147.7, 137.9, 133.2, 128.3, 121.2, 115.2, 98.9, 82.7, 38.4, 38.0, 28.6, 26.7, 25.1, 24.9, 24.7, 16.3. HRMS (ESI) calcd for $C_{18}H_{22}N_2O_6$ ([M + Na]$^+$) *m*/*z* 385.1370, found 385.1362.

*4.5.15. (R)-2,2-Dimethyl-1-phenylbut-3-en-1-ol (**31**).* Colorless oil in 97% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 90/10, 1.0 mL/min), t_{minor} = 5.150 min, t_{major} = 6.786 min; ee = 99%. $[\alpha]_D^{25}$ = +7.096 (*c* = 1.55, MeOH). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.25−7.16 (m, 5H), 5.84 (dd, *J* = 17.5, 10.8 Hz, 1H), 5.04 (dd, *J* = 38.2, 14.1 Hz, 2H), 4.35 (s, 1H), 1.91 (s, 1H), 0.94 (s, 3H), 0.89 (s, 3H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 144.0, 139.6, 126.7, 126.4, 126.3, 112.8, 79.5, 41.2, 23.4, 19.9. HRMS (ESI) calcd for $C_{12}H_{16}O$ ([M + H]$^+$) *m*/*z* 177.1274, found 177.1274.

*4.5.16. (1R,2R)-2,6-Dimethyl-1-phenyl-2-vinylhept-5-en-1-ol (**32**).* Colorless oil in 92% yield. Enantiomeric excess was determined by HPLC with a Chiralpak AD-H column (hexane/*i*PrOH = 98/2, 1.0 mL/min), t_{minor} = 9.570 min, t_{major} = 10.661 min; ee = 77%. $[\alpha]_D^{25}$ = +13.83 (*c* = 1.50, EtOH). The reported value[24] for the *S*,*S*-enantiomer (94% ee) is $[\alpha]_D^{24}$ = −12.32 (*c* = 1.16, EtOH). ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.29−7.23 (m, 5H), 5.84 (dd, *J* = 17.6, 10.8 Hz, 1H), 5.25 (dd, *J* = 10.8, 1.3 Hz, 1H), 5.11−5.00 (m, 2H), 4.39 (d, *J* = 5.8 Hz, 1H), 2.09 (s, 1H), 1.85 (dd, *J* = 16.2, 7.9 Hz, 2H), 1.65 (s, 3H), 1.54 (s, 3H), 1.44−1.22 (m, 2H), 0.91 (s, 3H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 143.9, 140.5, 131.2, 128.1, 127.5, 127.4, 124.7, 115.8, 80.0, 45.9, 37.5, 25.6, 22.8, 17.6, 16.2. HRMS (ESI) calcd for $C_{17}H_{24}O$ ([M + Na]$^+$) *m*/*z* 267.1713, found 267.1719.

*4.5.17. (1R,2S)-2,6-Dimethyl-1-phenyl-2-vinylhept-5-en-1-ol (**33**).* Colorless oil in 87% yield. Enantiomeric excess was determined by HPLC with a Chiralpak AD-H column (hexane/*i*PrOH = 98/2, 1.0 mL/min), t_{minor} = 9.793 min, t_{major}= 11.241 min; ee = 95%. $[\alpha]_D^{25}$ = −5.58 (*c* = 1.15, EtOH). The reported value[24] for the *S*,*R*-enantiomer (98% ee) is $[\alpha]_D^{24}$ = +4.33 (*c* = 1.05, EtOH). ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.25−7.19 (m, 5H), 5.72 (dd, *J* = 17.6, 10.9 Hz, 1H), 5.13 (d, *J* = 10.8 Hz, 1H), 4.95 (d, *J* = 17.5 Hz, 2H), 4.37 (d, *J* = 10.0 Hz, 1H), 1.80 (ddd, *J* = 28.6, 14.5, 7.1 Hz, 3H), 1.58 (s, 3H), 1.48 (s, 3H), 1.31−1.26 (m, 2H), 1.01 (s, 3H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 141.6, 140.1, 130.1, 126.8, 126.4, 126.3, 123.7, 114.2, 79.7, 44.2, 35.3, 24.6, 21.7, 17.9, 16.6. HRMS (ESI) calcd for $C_{17}H_{24}O$ ([M + Na]$^+$) *m*/*z* 267.1719, found 267.1713.

*4.5.18. (1R,2R)-1-Phenyl-2-vinylpentan-1-ol (**34**).* Colorless oil in 97% yield. Enantiomeric excess was determined by HPLC with a Chiralpak AD-H column (hexane/*i*PrOH = 98/2, 1.0 mL/min), t_{minor} = 10.884 min, t_{major} = 11.443 min; ee = 91%. $[\alpha]_D^{25}$ = +21.45 (*c* = 0.85, MeOH). The absolute configuration was determined by analogy.

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^{1}H NMR (600 MHz, $CDCl_3$) δ 7.28−7.19 (m, 5H), 5.62−5.55 (m, 1H), 5.20−5.08 (m, 2H), 4.31 (d, J = 7.8 Hz, 1H), 2.23 (dt, J = 13.6, 6.2 Hz, 1H), 2.11 (s, 1H), 1.27−1.05 (m, 4H), 0.71 (t, J = 7.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (151 MHz, $CDCl_3$) δ 142.5, 139.4, 128.2, 127.6, 127.0, 118.7, 52.5, 32.5, 20.3, 13.9. HRMS (ESI) calcd for $C_{13}H_{18}O$ ($[M + Na]^+$) m/z 213.1250, found 213.1241.

*4.5.19. (R)-((R)-Cyclohex-2-en-1-yl)(phenyl)methanol (**35**).* Colorless oil in 94% yield. Enantiomeric excess was determined by HPLC with a Chiralpak AD-H column (hexane/*i*PrOH = 90/10, 1.0 mL/min), t_{major} = 6.273 min, t_{minor} = 6.518 min; ee = 23%. $[\alpha]_D^{25}$ = +11.54 (c = 0.85, CH_2Cl_2). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.30−7.19 (m, 5H), 5.75 (ddd, J = 9.8, 6.1, 3.4 Hz, 1H), 5.32 (dd, J = 10.2, 1.9 Hz, 1H), 4.53 (d, J = 6.5 Hz, 1H), 2.44 (ddq, J = 11.3, 5.8, 2.7 Hz, 1H), 1.92 (t, J = 6.6 Hz, 2H), 1.70−1.62 (m, 2H), 1.53 (s, 1H), 1.46 (dt, J = 18.0, 6.2 Hz, 2H). $^{13}C\{^1H\}$ NMR (151 MHz, $CDCl_3$) δ 141.8, 129.4, 127.2, 126.9, 126.3, 125.4, 76.3, 41.9, 24.1, 22.7, 20.0. HRMS (ESI) calcd for $C_{13}H_{16}O$ ($[M + Na]^+$) m/z 211.1093, found 211.1098.

4.6. General Procedure for the Propargylation of Aldehydes. A screw-cap reaction tube loaded with a stir bar and 4 Å MS (50 mg) was evacuated, flame-dried, and back-filled with argon. To this tube were added the (*R*)-TRIP-PA catalyst **PA1** (5 mol %), freshly distilled aldehyde (0.1 mmol), and 1.0 mL of dry cyclohexane. Then, allenylboronic acid ester **B10** (0.12 mmol) was added via a syringe to the reaction mixture dropwise over 30 s. The reaction mixture was stirred for 8 h at room temperature and then directly loaded onto a silica gel column and was purified by flash chromatography using ethyl acetate and hexane (1:9).

*4.6.1. (R)-1-Phenyl-but-3-yn-1-ol (**36**).* Colorless oil in 94% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 95/5, 1.0 mL/min), t_{major} = 10.40 min, t_{minor} = 12.37 min; ee = 97%. $[\alpha]_D^{25}$ = +12.29 (c = 1.14, MeOH). The reported value[25] for the *R*-enantiomer (98% ee) is $[\alpha]_D^{24}$ = +12.9 (c = 1.55, MeOH). ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.43−7.28 (m, 5H), 4.88 (t, J = 6.4 Hz, 1H), 2.68−2.61 (m, 2H), 2.37 (s, 1H), 2.08 (t, J = 2.6 Hz, 1H). $^{13}C\{^1H\}$ NMR (151 MHz, $CDCl_3$) δ 141.4, 127.4, 127.0, 124.7, 79.6, 71.3, 69.9, 28.4. HRMS (ESI) calcd for $C_{10}H_{10}O$ ($[M + H]^+$) m/z 147.0804, found 147.0803.

*4.6.2. (R)-1-(4-Bromo-phenyl)-but-3-yn-1-ol (**37**).* Colorless oil in 94% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 98/2, 1.0 mL/min), t_{major} = 20.57 min, t_{minor} =23.23 min; ee = 99%. $[\alpha]_D^{25}$ = +21.89 (c = 1.21, $CHCl_3$). The reported value[26] for the *S*-enantiomer (81% ee) is $[\alpha]_D$ = −28.4 (c = 1, $CHCl_3$). ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.53−7.42 (m, 2H), 7.32−7.22 (m, 2H), 4.88−4.78 (m, 1H), 2.66−2.53 (m, 2H), 2.46 (s, 1H), 2.07 (t, J = 2.6 Hz, 1H). $^{13}C\{^1H\}$ NMR (151 MHz, $CDCl_3$) δ 140.3, 130.5, 126.4, 120.8, 79.1, 70.6, 70.3, 28.4. HRMS (ESI) calcd for $C_{10}H_9BrO$ ($[M + H]^+$) m/z 224.9910, found 224.9909.

*4.6.3. (R)-1-(4-Nitro-phenyl)-but-3-yn-1-ol (**38**).* Colorless oil in 93% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OJ-H column (hexane/*i*PrOH = 90/10, 1.0 mL/min), t_{minor} = 29.91 min, t_{major} =33.41 min; ee = 97%. $[\alpha]_D^{25}$ = +4.08 (c = 1.07, $CHCl_3$). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 8.22 (d, J = 8.7 Hz, 2H), 7.59 (d, J = 8.6 Hz, 2H), 5.00 (t, J = 5.8 Hz, 1H), 2.75−2.64 (m, 2H), 2.59 (s, 1H), 2.11 (t, J = 2.6 Hz, 1H). $^{13}C\{^1H\}$ NMR (151 MHz, $CDCl_3$) δ 148.4, 146.5, 125.6, 122.6, 78.3, 70.9, 70.2, 28.5. HRMS (ESI) calcd for $C_{10}H_9NO_3$ ($[M + H]^+$) m/z 192.0655, found 192.0654.

*4.6.4. (R)-1-(4-Methoxy-phenyl)-but-3-yn-1-ol (**39**).* Colorless oil in 91% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 90/10, 1.0 mL/min), t_{major} = 8.72 min, t_{minor} = 10.75 min; ee = 95%. $[\alpha]_D^{25}$ = +34.74 (c = 1.45, $CHCl_3$). The reported value[4g] for the *S*-enantiomer (89% ee) is $[\alpha]_D^{28}$ = −36.2 (c = 1, $CHCl_3$). ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.25 (d, J = 8.6 Hz, 2H), 6.82 (d, J = 8.7 Hz, 2H), 4.77 (t, J = 6.4 Hz, 1H), 3.74 (s, 3H), 2.60−2.52 (m, 2H), 2.00 (t, J = 2.6 Hz, 1H). $^{13}C\{^1H\}$ NMR (151 MHz, $CDCl_3$) δ 158.3, 133.6, 126.0, 112.8, 79.8, 70.9, 69.8, 54.2, 28.3. HRMS (ESI) calcd for $C_{11}H_{12}O_2$ ($[M + H]^+$) m/z 177.0910, found 177.0909.

*4.6.5. (R)-1-(3-Methoxy-phenyl)-but-3-yn-1-ol (**40**).* Colorless oil in 96% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 98/2, 1.0 mL/min), t_{major} = 36.16 min, t_{minor} = 41.49 min; ee = 96%. $[\alpha]_D^{25}$ = +7.39 (c = 0.59, $CHCl_3$). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.21 (d, J = 7.9 Hz, 1H), 6.92−6.86 (m, 2H), 6.77 (d, J = 7.2 Hz, 1H), 4.78 (t, J = 6.3 Hz, 1H), 3.74 (s, 3H), 2.57 (dd, J = 6.7, 2.7 Hz, 2H), 2.32 (s, 1H), 2.01 (t, J = 2.6 Hz, 1H). $^{13}C\{^1H\}$ NMR (151 MHz, $CDCl_3$) δ 158.7, 143.1, 128.5, 117.0, 112.4, 110.2, 79.6, 71.2, 69.9, 54.2, 28.4. HRMS (ESI) calcd for $C_{11}H_{12}O_2$ ($[M + H]^+$) m/z 177.0910, found 177.0909.

*4.6.6. (R)-1-o-Tolyl-but-3-yn-1-ol (**41**).* Colorless oil in 94% yield. Enantiomeric excess was determined by HPLC with a Chiralpak AD-H column (hexane/*i*PrOH = 98/2, 1.0 mL/min), t_{major} = 16.07 min, t_{minor} = 20.40 min; ee = 93%. $[\alpha]_D^{25}$ = +34.97 (c = 1.0, $CHCl_3$). The reported value[26] for the *S*-enantiomer (89% ee) is $[\alpha]_D^{25}$ = −63.2 (c = 0.58, $CHCl_3$). ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.44 (d, J = 7.5 Hz, 1H), 7.18−7.05 (m, 3H), 5.05 (dd, J = 7.7, 5.0 Hz, 1H), 2.58−2.49 (m, 2H), 2.29 (s, 3H), 2.02 (t, J = 2.6 Hz, 1H). $^{13}C\{^1H\}$ NMR (151 MHz, $CDCl_3$) δ 139.4, 133.5, 129.4, 126.7, 125.3, 124.0, 79.9, 69.7, 67.8, 27.2, 18.0. HRMS (ESI) calcd for $C_{11}H_{12}O$ ($[M + H]^+$) m/z 161.0961, found 161.0960.

*4.6.7. (R)-Methyl 4-(1-hydroxybut-3ynyl)benzoate (**42**).* Colorless oil in 96% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 90/10, 1.0 mL/min), t_{major} = 11.88 min, t_{minor} = 16.97 min; ee = 97%. $[\alpha]_D^{25}$ = +34.56 (c = 1.08, $CHCl_3$). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.95 (d, J = 8.3 Hz, 2H), 7.39 (d, J = 8.2 Hz, 2H), 4.92−4.78 (m, 1H), 3.84 (s, 3H), 2.62−2.53 (m, 2H), 2.37 (s, 1H), 2.01 (t, J = 2.6 Hz, 1H). $^{13}C\{^1H\}$ NMR (151 MHz, $CDCl_3$) δ 165.8, 146.3, 128.7, 124.7, 79.0, 70.8, 70.4, 51.1, 28.4. HRMS (ESI) calcd for $C_{12}H_{12}O_3$ ($[M + H]^+$) m/z 205.0859, found 205.0860.

*4.6.8. (R)-1-(Benzo[d][1,3]dioxol-5-yl)but-3-yn-1-ol (**43**).* Colorless oil in 94% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 95/5, 1.0 mL/min), t_{major} = 15.89 min, t_{minor} = 22.40 min; ee = 95%. $[\alpha]_D^{25}$ = +3.89 (c = 1.49, $CHCl_3$). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 6.77 (ddd, J = 35.9, 35.2, 4.7 Hz, 3H), 5.89 (s, 2H), 4.73 (t, J = 6.4 Hz, 1H), 2.57−2.50 (m, 2H), 2.24 (s, 1H), 2.01 (t, J = 2.6 Hz, 1H). $^{13}C\{^1H\}$ NMR (151 MHz, $CDCl_3$) δ 146.7, 146.2, 135.5, 118.2, 107.1, 105.2, 100.0, 79.5, 71.2, 70.0, 28.5. HRMS (ESI) calcd for $C_{11}H_{10}O_3$ ($[M + H]^+$) m/z 191.0703, found 191.0701.

*4.6.9. (R)-1-Thiophen-2-yl-but-3-yn-1-ol (**44**).* Colorless oil in 95% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OJ-H column (hexane/*i*PrOH = 90/10, 1.0 mL/min), t_{minor} = 16.64 min, t_{major} = 18.13 min; ee = 91%. $[\alpha]_D^{25}$ = −19.42 (c = 1.54, EtOH). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.27 (d, J = 6.1 Hz, 1H), 7.07−6.93 (m, 2H), 5.17−5.07 (m, 1H), 2.78−2.74 (m, 2H), 2.49 (d, J = 3.2 Hz, 1H), 2.11 (t, J = 2.6 Hz, 1H). $^{13}C\{^1H\}$ NMR (151 MHz, $CDCl_3$) δ 145.1, 125.6, 123.9, 123.1, 79.0, 70.5, 67.5, 28.5. HRMS (ESI) calcd for C_8H_8OS ($[M + H]^+$) m/z 153.0369, found 153.0367.

*4.6.10. (R,E)-1-Phenylhex-1-en-5-yn-3-ol (**45**).* Colorless oil in 94% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OD-H column (hexane/*i*PrOH = 90/10, 1.0 mL/min), t_{major} = 9.51 min, t_{minor} = 13.17 min; ee = 91%. $[\alpha]_D^{25}$ = −49.32 (c = 0.62, benzene). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.32 (d, J = 7.3 Hz, 2H), 7.25 (t, J = 7.6 Hz, 2H), 7.18 (d, J = 8.5 Hz, 1H), 6.59 (d, J = 15.9 Hz, 1H), 6.21 (dd, J = 15.9, 6.3 Hz, 1H), 4.41 (q, J = 5.9 Hz, 1H), 2.54−2.43 (m, 2H), 2.08 (s, 1H), 2.02 (t, J = 2.6 Hz, 1H). $^{13}C\{^1H\}$ NMR (151 MHz, $CDCl_3$) δ 135.3, 130.3, 128.9, 127.5, 127.1, 126.8, 125.5, 79.1, 70.1, 69.7, 26.7. HRMS (ESI) calcd for $C_{12}H_{12}O$ ($[M + H]^+$) m/z 173.0961, found 173.0961.

4.7. Procedures for Synthetic Application. Compound **47** was prepared following the literature procedure[15b] with starting alcohol **19** with 99% ee, dr > 20:1.

*4.7.1. (1S,2R)-1-(4-Bromophenyl)-2-methylbut-3-en-1-yl(S)-2-((tert-butoxycarbonyl)amino)-3-hydroxy-3-methylbutanoate (**47**).*

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The title compound was obtained by flash column chromatography (hexane/EA, 8: 1) as a colorless oil in 86% yield. Enantiomeric excess was determined by HPLC with a Chiralpak IF column (hexane/*i*PrOH = 80/20, 1.0 mL/min), t_{minor} = 7.233 min, t_{major} = 8.502 min; ee = 99%. $[\alpha]_D^{25}$ = −2.54 (*c* = 0.75, MeOH). The absolute configuration was determined by analogy. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.39−7.35 (m, 2H), 7.10 (t, *J* = 8.1 Hz, 2H), 5.64−5.43 (m, 2H), 5.31−5.20 (m, 1H), 5.02−4.83 (m, 2H), 4.21−4.07 (m, 1H), 2.59 (ddd, *J* = 86.5, 47.1, 31.8 Hz, 2H), 1.36 (d, *J* = 14.0 Hz, 9H), 1.18 (d, *J* = 7.1 Hz, 6H), 1.03−0.97 (m, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 171.4, 155.6, 138.3, 137.1, 131.3, 129.0, 122.1, 116.5, 80.1, 79.6, 72.2, 71.8, 61.1, 42.6, 28.2, 27.0, 26.1, 21.6. HRMS (ESI) calcd for $C_{21}H_{30}BrNO_5$ ($[M + Na]^+$) *m*/*z* 478.1200, found 478.1204.

Compounds **49** and **50** were prepared following the literature procedure[27] with starting alcohol **19** with 89% ee, dr > 20:1.

*4.7.2. (1R,2R)-1-(4-Bromophenyl)-2-methylbut-3-en-1-yl acrylate (**49**).* The title compound was obtained by flash column chromatography (hexane/EA = 10:1) as a colorless oil in 92% yield. Enantiomeric excess was determined by HPLC with a Chiralcel OJ-H column (hexane/*i*PrOH = 98/2, 1.0 mL/min), t_{major} = 5.293 min, t_{minor} = 8.161 min; ee = 87%. $[\alpha]_D^{25}$ = +28.02 (*c* = 0.85, MeOH). The absolute configuration was determined by analogy. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.40−7.36 (m, 2H), 7.14−7.10 (m, 2H), 6.33 (dd, *J* = 17.3, 1.4 Hz, 1H), 6.06 (dd, *J* = 17.3, 10.4 Hz, 1H), 5.75 (dd, *J* = 10.4, 1.4 Hz, 1H), 5.72−5.62 (m, 1H), 5.55 (d, *J* = 7.5 Hz, 1H), 4.99−4.93 (m, 2H), 2.60 (dt, *J* = 14.5, 7.2 Hz, 1H), 0.85 (d, *J* = 6.9 Hz, 3H). ^{13}C{^{1}H} NMR (151 MHz, $CDCl_3$) δ 164.1, 138.0, 136.9, 130.3, 130.0, 127.7, 127.3, 120.8, 115.0, 77.5, 42.3, 15.1. HRMS (ESI) calcd for $C_{14}H_{15}BrO_2$ ($[M + Na]^+$) *m*/*z* 317.0148, found 317.0144.

*4.7.3. (5R,6R)-6-(4-Bromophenyl)-5-methyl-5,6-dihydro-2H-pyran-2-one (**50**).* The title compound was obtained by flash column chromatography (hexane/EA = 2:1) as a white solid in 84% yield. mp = 109−110 °C. Enantiomeric excess was determined by HPLC with a Chiralpak AD-H column (hexane/*i*PrOH = 75/25, 1.0 mL/min), t_{major} = 6.826 min, t_{minor} = 7.334 min; ee = 82%. $[\alpha]_D^{25}$ = +92.41 (*c* = 1.25, MeOH). The absolute configuration was determined by analogy. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.58−7.48 (m, 2H), 7.25 (d, *J* = 1.8 Hz, 2H), 6.74 (dd, *J* = 9.8, 2.1 Hz, 1H), 6.08 (dd, *J* = 9.8, 2.6 Hz, 1H), 4.94 (d, *J* = 10.8 Hz, 1H), 2.83−2.71 (m, 1H), 1.01 (d, *J* = 7.3 Hz, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 162.6, 150.0, 135.3, 130.7, 128.0, 122.0, 119.4, 84.2, 34.2, 14.7. HRMS (ESI) calcd for $C_{12}H_{11}BrO_2$ ($[M + Na]^+$) *m*/*z* 288.9835, found 288.9833.

4.8. General Procedure for Gram-Scale Reaction. The reaction was conducted on a 10.0 mmol scale of benzaldehyde according to the above general allylboration method. Boronate **B2**, **B3**, and **B10** were added to the reaction system over 2 h. The product was purified by silica gel flash column chromatography (ethyl acetate/hexane = 1:9) to afford desired product **51** (1.43 g, 97% yield, 99% ee), **52** (2.13 g, 97% yield, 93% ee, dr > 20:1), and **53** (1.41 g, 97% yield, 98% ee) as a colorless oil.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.0c01646.

Experimental details; experimental preparations; characterization; NMR spectra; chiral HPLC conditions; and other data (PDF)

Crystallographic data for **30** (CIF)

CheckCIF/PLATON report (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Jon C. Antilla − *Institute for Molecular Design and Synthesis, School of Pharmaceutical Science and Technology, Tianjin University, Tianjin 300072, China; School of Sciences, Zhejiang Sci-Tech University, Hangzhou City, Zhejiang Province 310018, China*; Email: jantilla@tju.edu.cn

Authors

Jinping Yuan − *Institute for Molecular Design and Synthesis, School of Pharmaceutical Science and Technology, Tianjin University, Tianjin 300072, China*

Pankaj Jain − *Novartis Institutes for BioMedical Research, Cambridge, Massachusetts 02139, United States*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.joc.0c01646

Notes

The authors declare no competing financial interest.

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